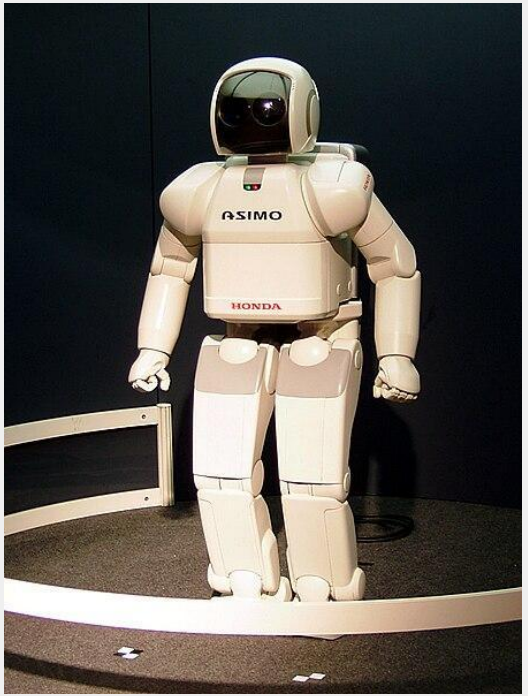
History of AI

Global Perspectives in History of AI



unsojo@cornell.edu

Lecture 1 January 24, 2025

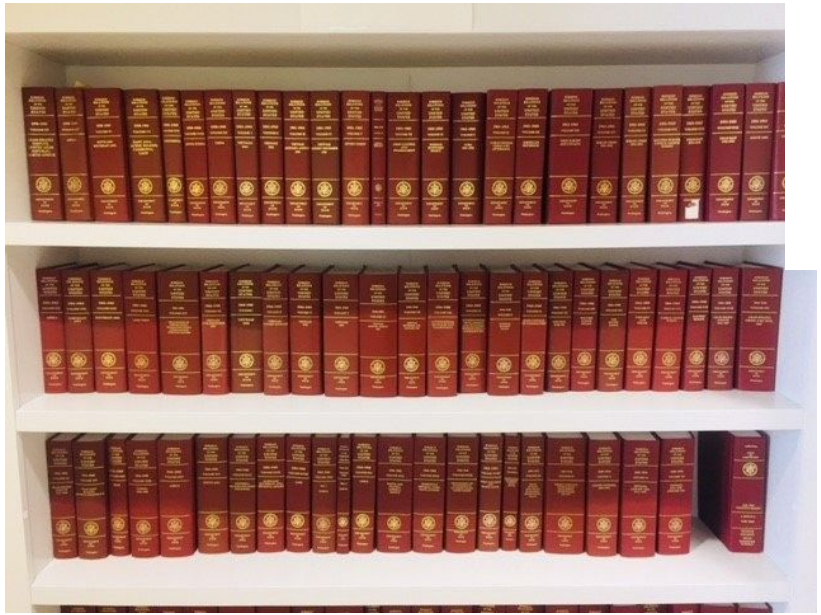
Nice to Meet You!

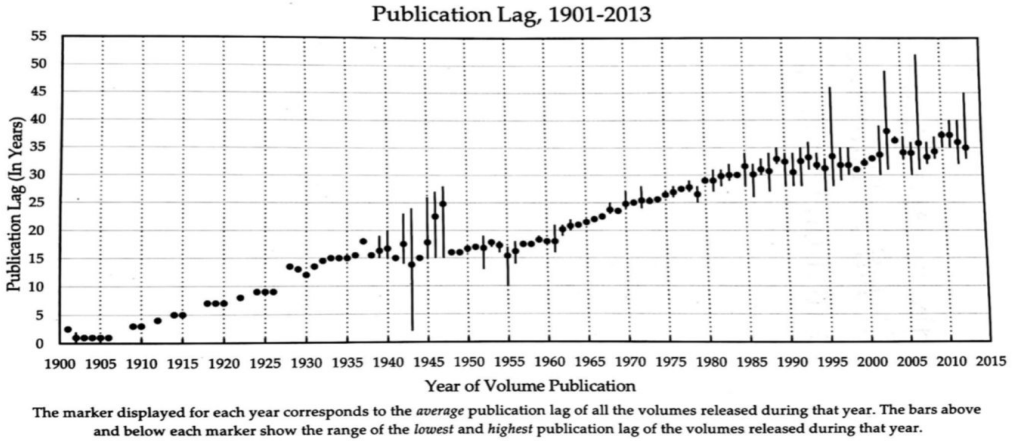
Unso is pronounced however you want, but easiest to remember as part of:
UNSOLVED, **UNSO**LICITED, **UNSO**PHISTICATED





FRUS as an Archive





The marker displayed for each year corresponds to the *average* publication lag of all the volumes released during that year. The bars above and below each marker show the range of the *lowest* and *highest* publication lag of the volumes released during that year.

Document Type	Count	Percentile
Unidentified Correspondences	249,786	81.655
Telegrams	21,106	6.9
Memoranda	18,102	5.919
Letters	3,454	1.129
Non-correspondences	3,471	1.134

Table 2.2: The distribution of types of documents in the corpus. Most documents are correspondences and the document type was identified only if specified in the document header. All other correspondence type documents fall under 'Unidentified Correspondences' which always includes sender and recipient information. In practice most of the documents under this category are telegrams. Other document types not mentioned include cablegrams, notes, messages, airgrams, and editorials, all of which amount to less than 3% of the documents.

The OBJECTIVE of this course is...

1. Understand that technology (such as AI is an enmeshed part of society)
2. See how the World is interconnected!
3. Appreciate the big picture.
4. Answer: what is AI??

This class is like a humanities class for techies/techie class for fuzzies?

1. Discussion-driven: Do your readings before Fridayyy

2. Writing-heavy: But I don't judge your English or Prose style

3. Push you to think critically: Be open-minded!

What will we talk about?

- Early modern conceptions of machinery/man-made living things
- What is AI anyway?
- Is open-source a recent phenomenon?
- Ethical problems: policing, hiring, surveillance
- How have people been fantasizing about AI?
- Is AI just deep learning?
- Software vs. hardware constraints
- Isn’t AI an American invention? Who will dominate AI?
- Do all domains adopt AI the same?
- How does a new field take-off? Story of a modern discipline
- Difference between AI, big data, machine learning, data science …

You will be graded on:

1. Group/Individual Class Presentation (30 mins-ish)

2. Midterm Argument Paper (week 10)
[fewer credits, shorter paper]

3. Final Research Paper
[fewer credits, shorter paper]

Intros:

1. Who you are
2. Why are you taking this course
3. Your favorite AI



TEAMS!

Why AI?

Trump Announces $100 Billion A.I. Initiative

OpenAI, Oracle and SoftBank formed a new Stargate to invest in data centers, building o investments in the technology.

Share full article



Sam Altman, chief executive of OpenAI, center, at the inauguration of President Trump at the U.S. Capitol on Monday. Pool photo by Al Drago

The Stargate Project is a new company which intends to invest $500 billion over the next four years building new AI infrastructure for OpenAI in the United States. We will begin deploying $100 billion immediately. This infrastructure will secure American leadership in AI, create hundreds of thousands of American jobs, and generate massive economic benefit for the entire world. This project will not only support the re-industrialization of the United States but also provide a strategic capability to protect the national security of America and its allies.

The initial equity funders in Stargate are SoftBank, OpenAI, Oracle, and MGX. SoftBank and OpenAI are the lead partners for Stargate, with SoftBank having financial responsibility and OpenAI having operational responsibility. Masayoshi Son will be the chairman.

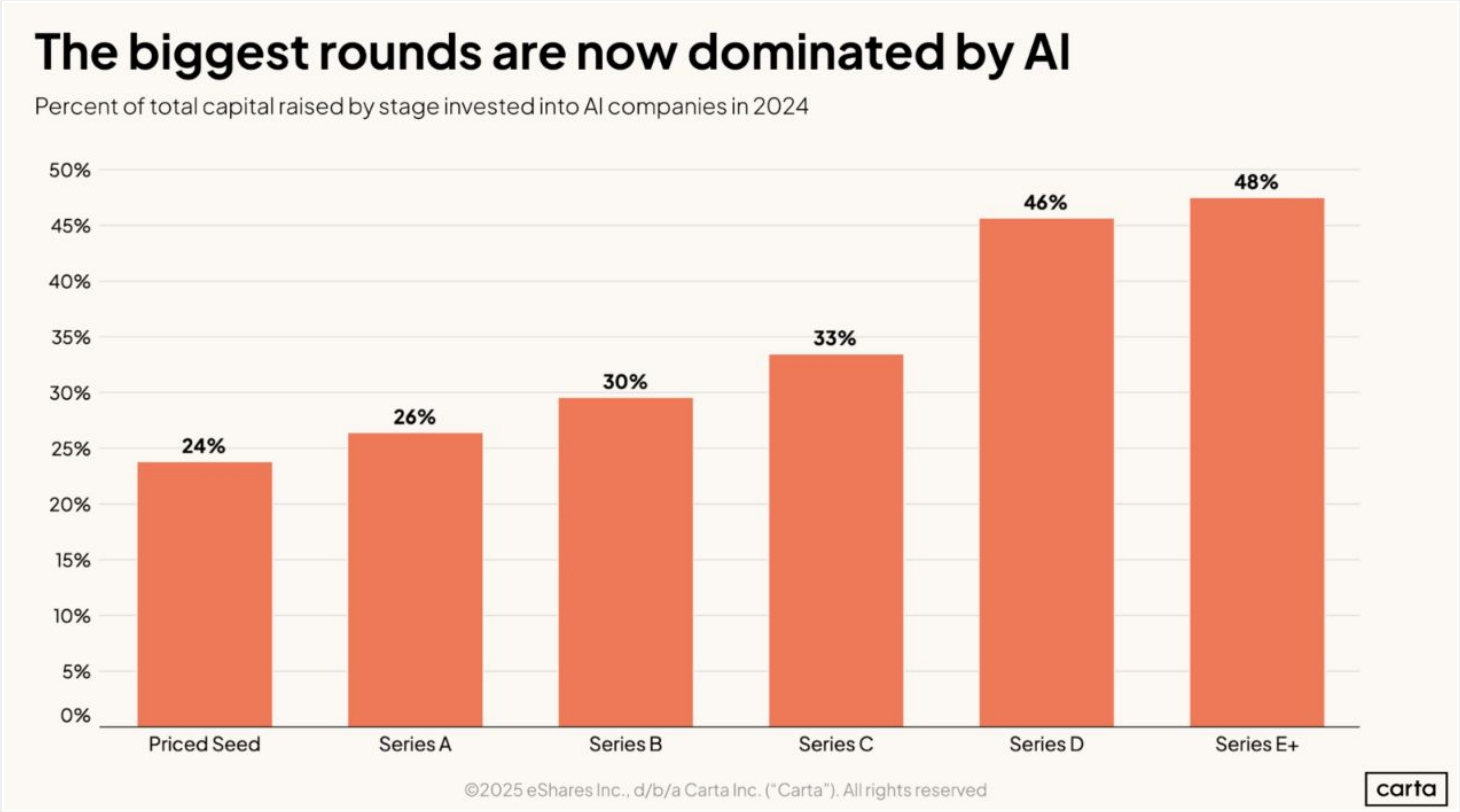
Arm, Microsoft, NVIDIA, Oracle, and OpenAI are the key initial technology partners. The buildout is currently underway, starting in Texas, and we are evaluating potential sites across the country for more campuses as we finalize definitive agreements.

As part of Stargate, Oracle, NVIDIA, and OpenAI will closely collaborate to build and operate this computing system. This builds on a deep collaboration between OpenAI and NVIDIA going back to 2016 and a newer partnership between OpenAI and Oracle.

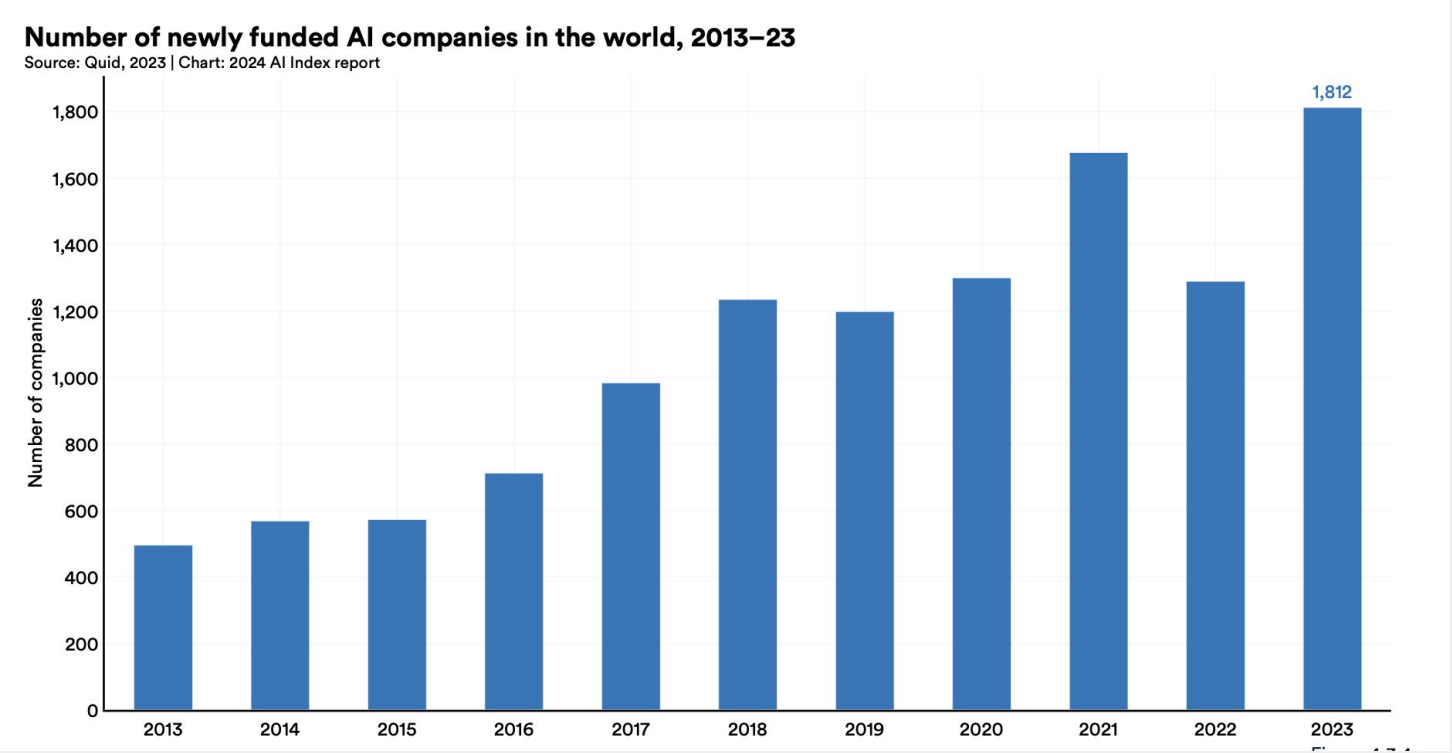
This also builds on the existing OpenAI partnership with Microsoft. OpenAI will continue to increase its consumption of Azure as OpenAI continues its work with Microsoft with this additional compute to train leading models and deliver great products and services.

All of us look forward to continuing to build and develop AI—and in particular AGI—for the benefit of all of humanity. We believe that this new step is critical on the path, and will enable creative people to figure out how to use AI to elevate humanity.

Why AI?

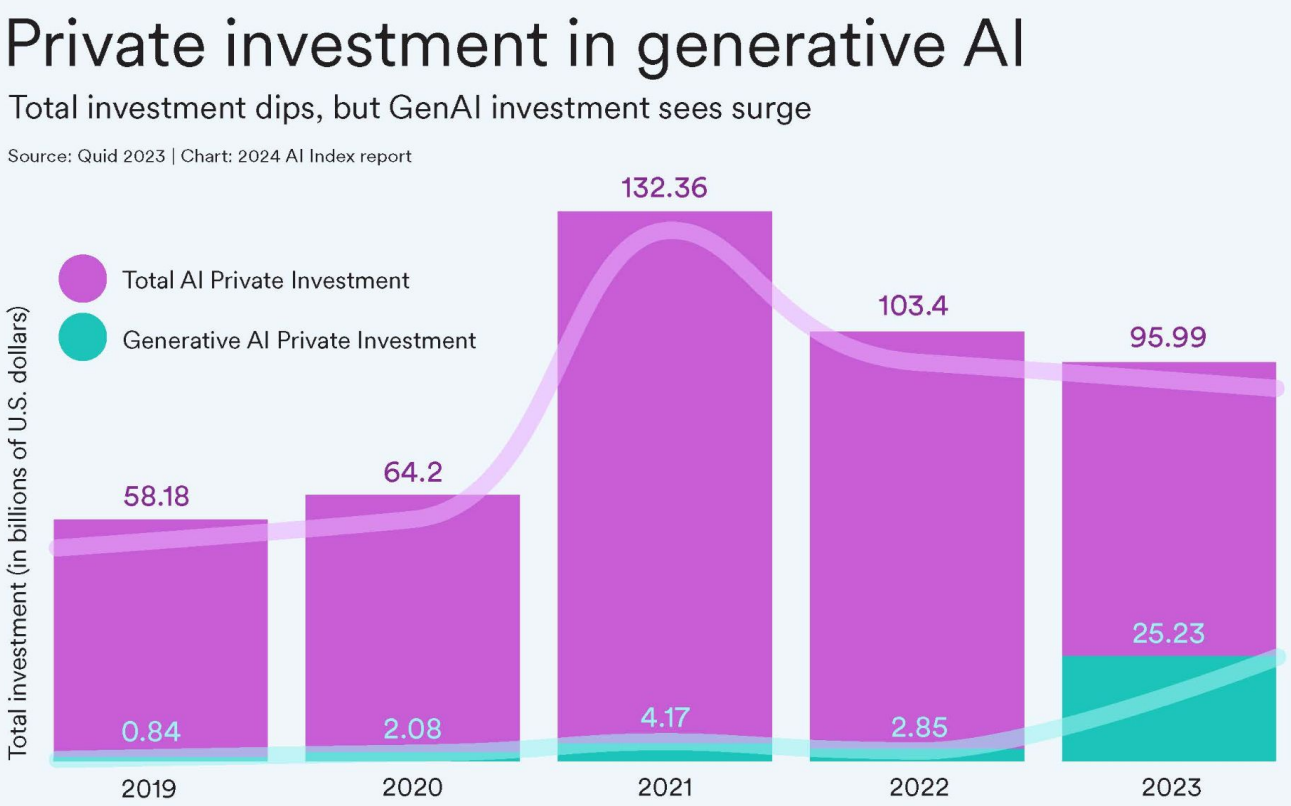

The biggest rounds are now dominated by AI
Percent of total capital raised by stage invested into AI companies in 2024
50%
45%
40%
35%
30%
25%
20%
15%
10%
5%
0%
24%
26%
30%
33%
46%
48%
Priced Seed
Series A
Series B
Series C
Series D
Series E+
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carta

Why AI?



AI Index (Stanford HAI) 2024

Why AI?

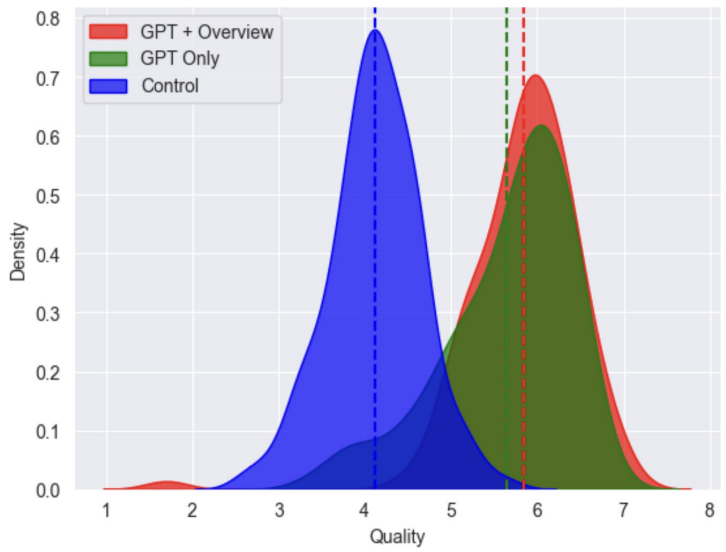


AI Index (Stanford HAI) 2024

Why AI?

BCG Consultants

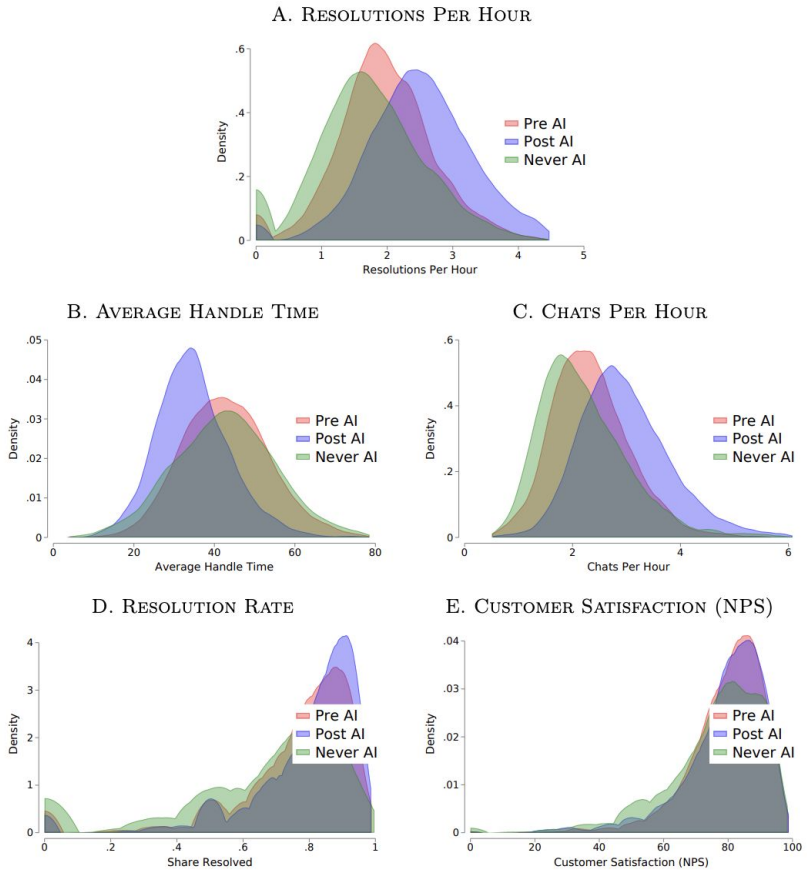
Figure 2: **Performance Distribution - Inside the Frontier**



Notes: This figure displays the full distribution of performance in the experimental task inside the frontier for subjects in the three experimental groups (red for subjects in the GPT+Overview condition; green for subjects in the GPT Only condition; blue for subjects in the control condition).

Customer Support Agents

FIGURE 2: RAW PRODUCTIVITY DISTRIBUTIONS, BY AI TREATMENT



Why AI?

Exhibit 4

Generative AI use cases will have different impacts on business functions across industries.

< Prev 01 — 02 Next >

Generative AI productivity impact by business functions[1]

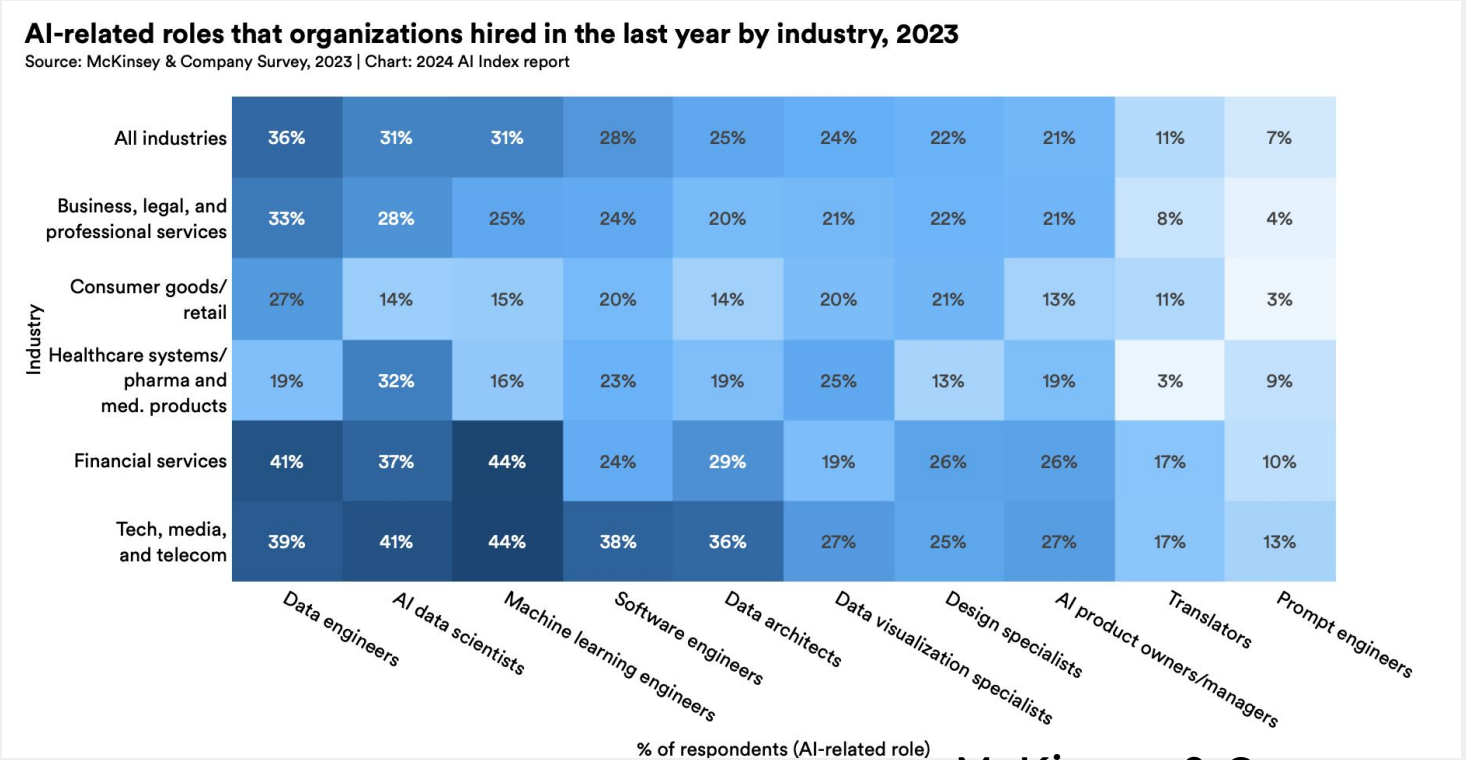
Low impact — High impact

	Total, % of industry revenue	**Total,** $ billion	Marketing and sales	Customer operations	Product R&D	Software engineering	Supply chain and operations	Risk and legal	Strategy and finance	Corporate IT[2]	Talent and organization
			760–1,200	340–470	230–420	580–1,200	290–550	180–260	120–260	40–50	60–90
High tech	**4.8–9.3**	240–460									
Banking	**2.8–4.7**	200–340									
Pharmaceuticals and medical products	**2.6–4.5**	60–110									
Education	**2.2–4.0**	120–230									
Telecommunications	**2.3–3.7**	60–100									
Healthcare	**1.8–3.2**	150–260									
Insurance	**1.8–2.8**	50–70									
Media and entertainment	**1.8–3.1**	80–130									
Advanced manufacturing[3]	**1.4–2.4**	170–290									
Consumer packaged goods	**1.4–2.3**	160–270									
Advanced electronics and semiconductors	**1.3–2.3**	100–170									

Note: Figures may not sum to 100%, because of rounding. [1]Excludes implementation costs (eg, training, licenses). [2]Excluding software engineering. [3]Includes aerospace, defense, and auto manufacturing. [4]Including auto retail.
Source: Comparative Industry Service (CIS), IHS Markit; Oxford Economics; McKinsey Corporate and Business Functions database; McKinsey Manufacturing and Supply Chain 360; McKinsey Sales Navigator; Ignite, a McKinsey database; McKinsey analysis

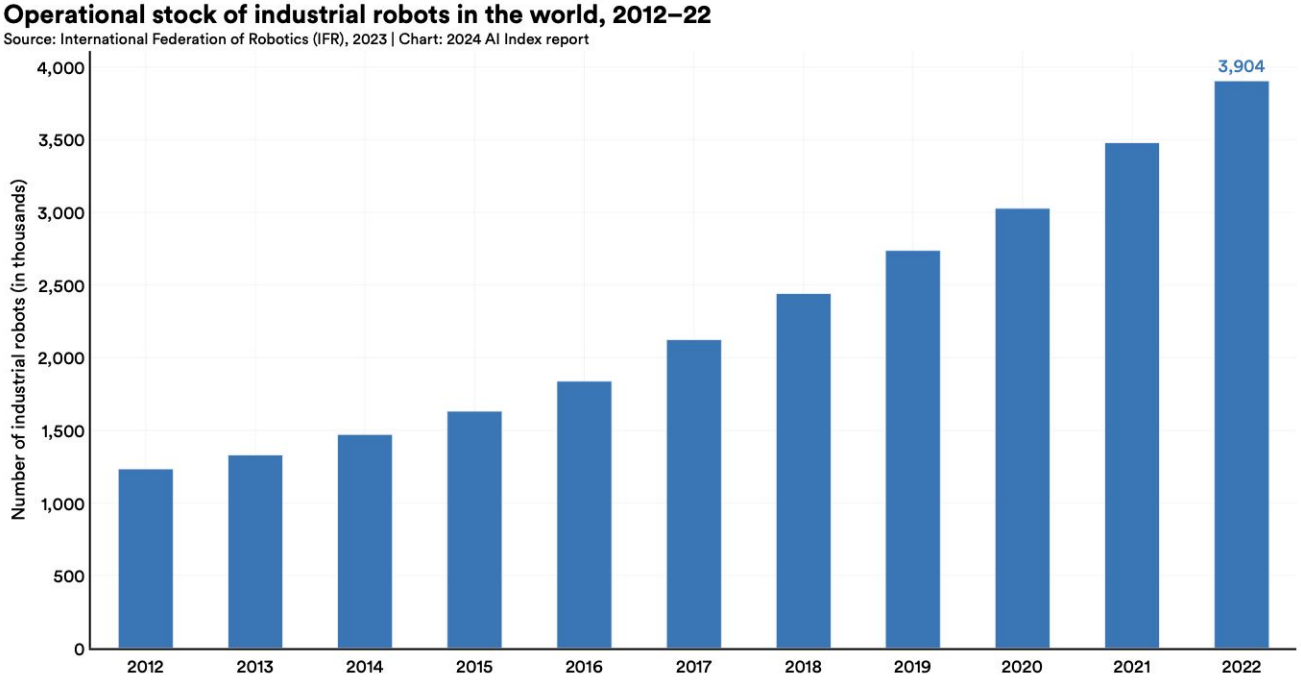
McKinsey & Company

Why AI?



McKinsey & Company

Why AI?


Operational stock of industrial robots in the world, 2012–22
Source: International Federation of Robotics (IFR), 2023 | Chart: 2024 AI Index report
Number of industrial robots (in thousands)
4,000
3,500
3,000
2,500
2,000
1,500
1,000
500
0
3,904
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
Figure 4.5.2

Why AI?

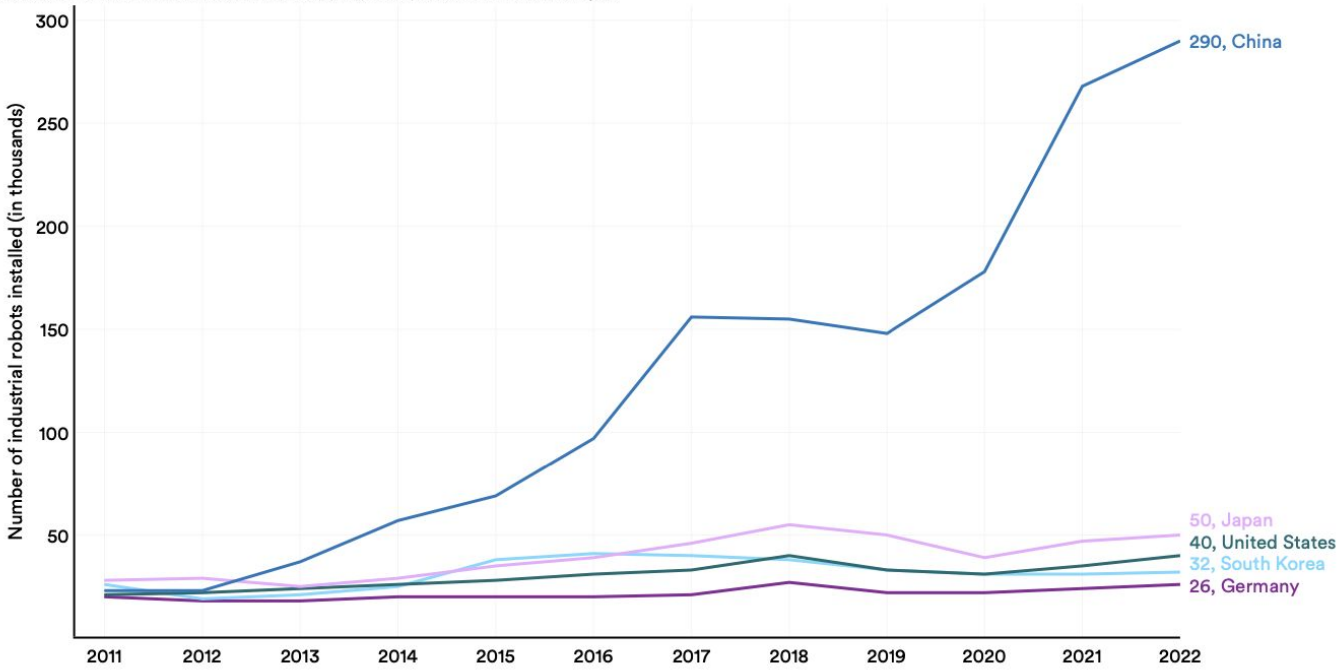


Figure 4.5.5

Why AI?

Original

```
Memory[0] = A
Memory[1] = B
Memory[2] = C

mov Memory[0] P  // P = A
mov Memory[1] Q  // Q = B
mov Memory[2] R  // R = C

mov R S
cmp P R
cmovg P R  // R = max(A, C)
cmovl P S  // S = min(A, C)
mov S P    // P = min(A, C)
cmp S Q
cmovg Q P // P = min(A, B, C)
cmovg S Q // Q = max(min(A, C), B)

mov P Memory[0]  // = min(A, B, C)
mov Q Memory[1]  // = max(min(A, C), B)
mov R Memory[2]  // = max(A, C)
```

AlphaDev

```
Memory[0] = A
Memory[1] = B
Memory[2] = C

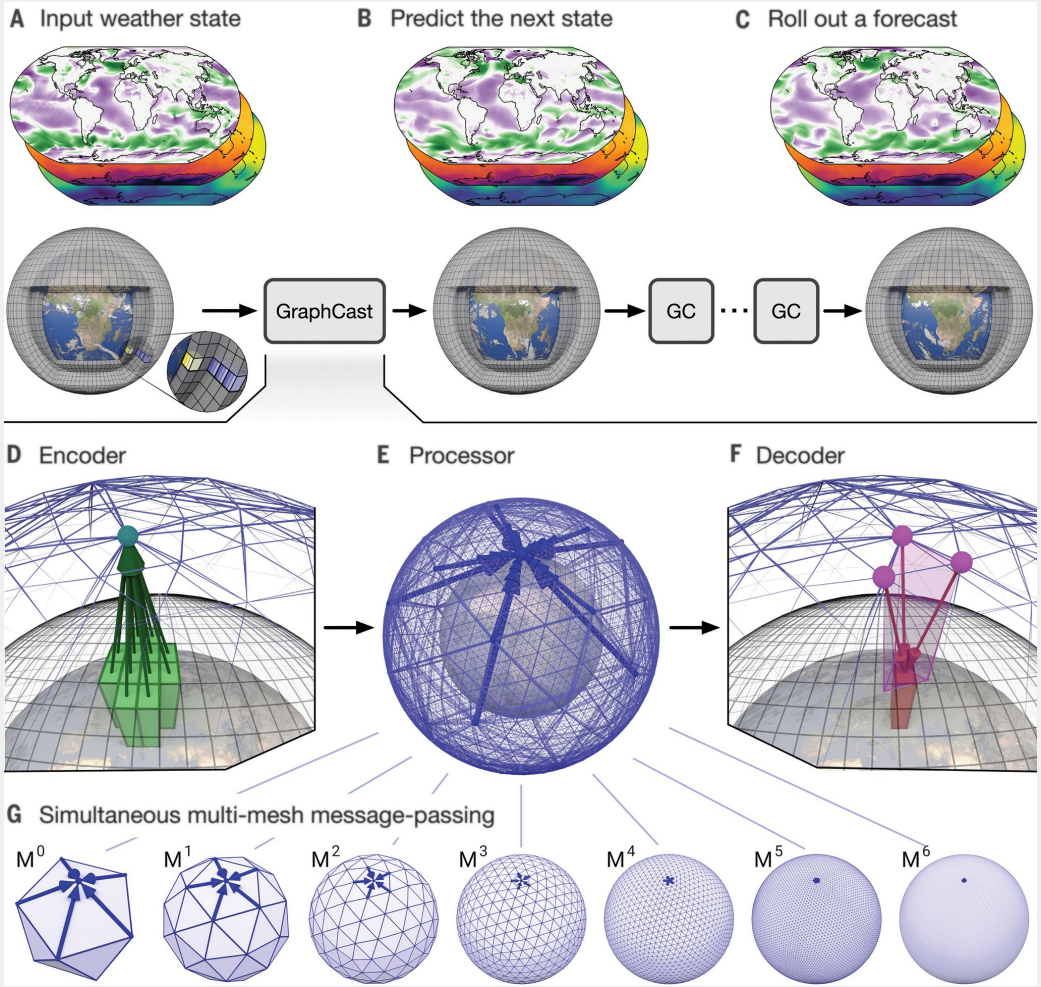
mov Memory[0] P  // P = A
mov Memory[1] Q  // Q = B
mov Memory[2] R  // R = C

mov R S
cmp P R
cmovg P R  // R = max(A, C)
cmovl P S  // S = min(A, C)

cmp S Q
cmovg Q P  // P = min(A, B)
cmovg S Q  // Q = max(min(A, C), B)

mov P Memory[0]  // = min(A, B)
mov Q Memory[1]  // = max(min(A, C), B)
mov R Memory[2]  // = max(A, C)
```

AlphaDev



GraphCast

Why AI?

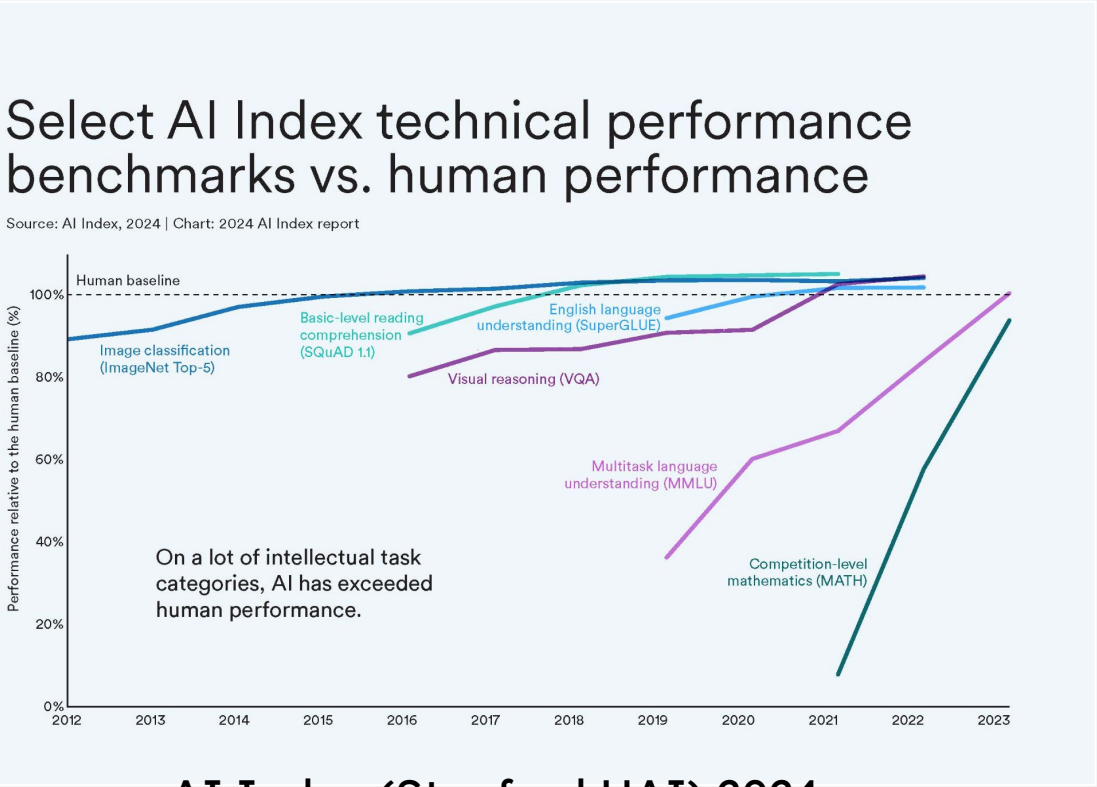
MATH Dataset (Ours)

Problem: Tom has a red marble, a green marble, a blue marble, and three identical yellow marbles. How many different groups of two marbles can Tom choose?

Solution: There are two cases here: either Tom chooses two yellow marbles (1 result), or he chooses two marbles of different colors ($\binom{4}{2} = 6$ results). The total number of distinct pairs of marbles Tom can choose is $1 + 6 = \boxed{7}$.

Problem: The equation $x^2 + 2x = i$ has two complex solutions. Determine the product of their real parts.

Solution: Complete the square by adding 1 to each side. Then $(x+1)^2 = 1 + i = e^{\frac{i\pi}{4}}\sqrt{2}$, so $x + 1 = \pm e^{\frac{i\pi}{8}}\sqrt[4]{2}$. The desired product is then $\left(-1 + \cos\left(\frac{\pi}{8}\right)\sqrt[4]{2}\right)\left(-1 - \cos\left(\frac{\pi}{8}\right)\sqrt[4]{2}\right) = 1 - \cos^2\left(\frac{\pi}{8}\right)\sqrt{2} = 1 - \frac{(1+\cos(\frac{\pi}{4}))}{2}\sqrt{2} = \boxed{\frac{1-\sqrt{2}}{2}}$.



AI Index (Stanford HAI) 2024

Why AI?

Simulated exams	GPT-4 estimated percentile	GPT-4 (no vision) estimated percentile	GPT-3.5 estimated percentile
Uniform Bar Exam (MBE+MEE+MPT)[1]	298/400 ~90th	298/400 ~90th	213/400 ~10th
LSAT	163 ~88th	161 ~83rd	149 ~40th
SAT Evidence-Based Reading & Writing	710/800 ~93rd	710/800 ~93rd	670/800 ~87th
SAT Math	700/800 ~89th	690/800 ~89th	590/800 ~70th
Graduate Record Examination (GRE) Quantitative	163/170 ~80th	157/170 ~62nd	147/170 ~25th
Graduate Record Examination (GRE) Verbal	169/170 ~99th	165/170 ~96th	154/170 ~63rd
Graduate Record Examination (GRE) Writing	4/6 ~54th	4/6 ~54th	4/6 ~54th
USABO Semifinal Exam 2020	87/150 99th–100th	87/150 99th–100th	43/150 31st–33rd
USNCO Local Section Exam 2022	36/60	38/60	24/60
Medical Knowledge Self-Assessment Program	75%	75%	53%
Codeforces Rating	392 below 5th	392 below 5th	260 below 5th
AP Art History	5 86th–100th	5 86th–100th	5 86th–100th
AP Biology	5 85th–100th	5 85th–100th	4 62nd–85th
AP Calculus BC	4 43rd–59th	4 43rd–59th	1 0th–7th
AP Chemistry	4 71st–88th	4 71st–88th	2 22nd–46th
AP English Language and Composition	2 14th–44th	2 14th–44th	2 14th–44th
AP English Literature and Composition	2 8th–22nd	2 8th–22nd	2 8th–22nd
AP Environmental Science	5 91st–100th	5 91st–100th	5 91st–100th
AP Macroeconomics	5 84th–100th	5 84th–100th	2 33rd–48th
AP Microeconomics	5 82nd–100th	4 60th–82nd	4 60th–82nd
AP Physics 2	4 66th–84th	4 66th–84th	3 30th–66th
AP Psychology	5 83rd–100th	5 83rd–100th	5 83rd–100th
AP Statistics	5 85th–100th	5 85th–100th	3 40th–63rd
AP US Government	5 88th–100th	5 88th–100th	4 77th–88th
AP US History	5 89th–100th	4 74th–89th	4 74th–89th
AP World History	4 65th–87th	4 65th–87th	4 65th–87th
AMC 10	30/150 6th–12th	36/150 10th–19th	36/150 10th–19th
AMC 12	60/150 45th–66th	48/150 19th–40th	30/150 4th–8th
Intro Sommelier (theory knowledge)	92%	92%	80%
Certified Sommelier (theory knowledge)	86%	86%	58%
Advanced Sommelier (theory knowledge)	77%	77%	46%
Leetcode (easy)	31/41	31/41	12/41
Leetcode (medium)	21/80	21/80	8/80
Leetcode (hard)	3/45	3/45	0/45

OpenAI 2024

Why AI?

THE U.S. Sun | News | Sport | Lifestyle | Entertainment | Money | Tech

A FORMER Google engineer who was let go by the tech giant and claimed its AI was sentient has made some more concerning claims.

Blake Lemoine made headlines last year after claiming Google's Large Language Model (LLM) had thoughts and feelings like a child.



Blake Lemoine is an AI consultant and public speaker who thinks some artificial intelligence could be sentient Credit: Getty

Google strongly denied these claims and said the AI was designed to respond like a human, but it's not sentient like us.

Lemoine has now published a new essay in Newsweek in which he discusses Microsoft's Bing's Open AI-powered chatbot.

The New York Times Account

A.I. Is Not Sentient. Why Do People Say It Is?

Robots can't think or feel, despite what the researchers who build them want to believe.

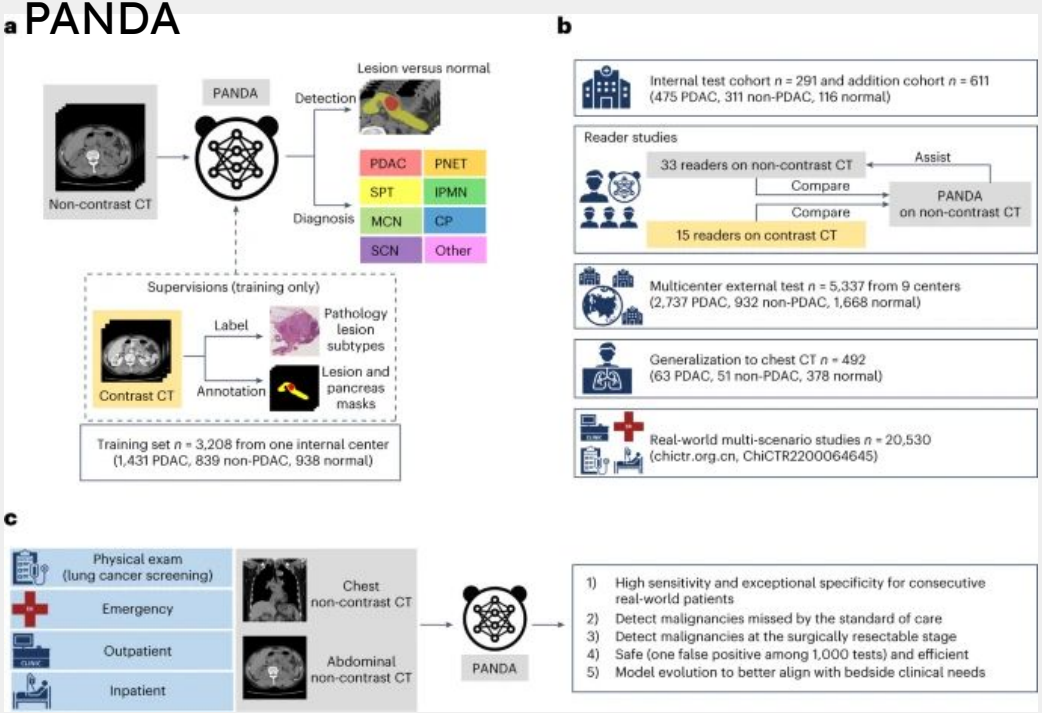


- How do you use AI?
- What changes or improvements have you seen in AI in your lifetime?
- How has AI changed your life?

What is AI anyway?



Types of AI: Narrow vs. General



Anthropic CEO says AGI is a marketing term and the next AI milestone will be like a 'country of geniuses in a data center'

Sarah Perkel Jan 22, 2025, 5:13 AM EST Share Save



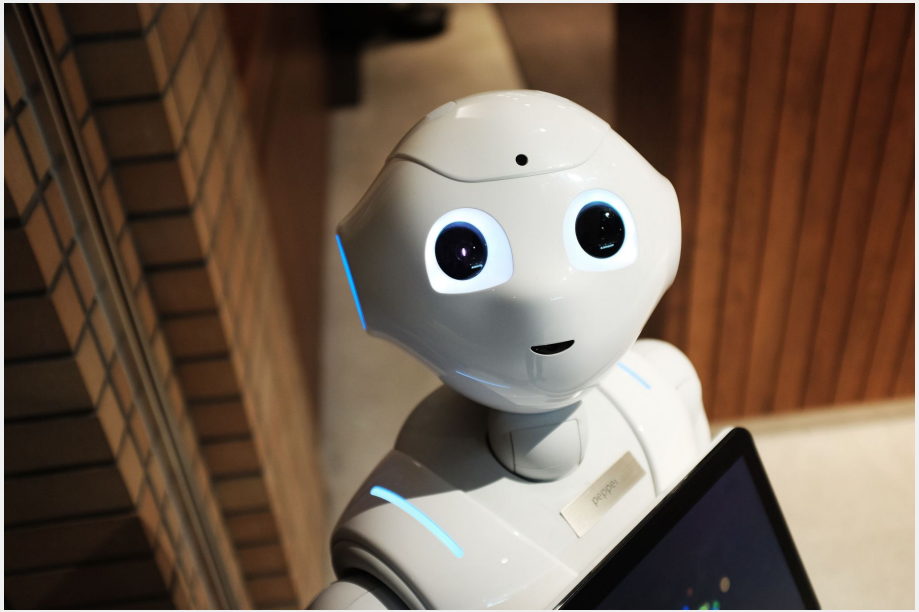
Anthropic CEO Dario Amodei said his preferred phrase to describe advanced AI was an "evocative phrase for all the power and all the positive things" and "all of the potential negative things."
Chesnot/Getty Images

- **There's no single definition for artificial general intelligence.**
- **Dario Amodei, the CEO of Anthropic, previously predicted that "powerful AI" could arrive by 2026.**
- **Amodei recently told CNBC that AGI was more of a "marketing term."**

Types of AI: System vs. Embodied

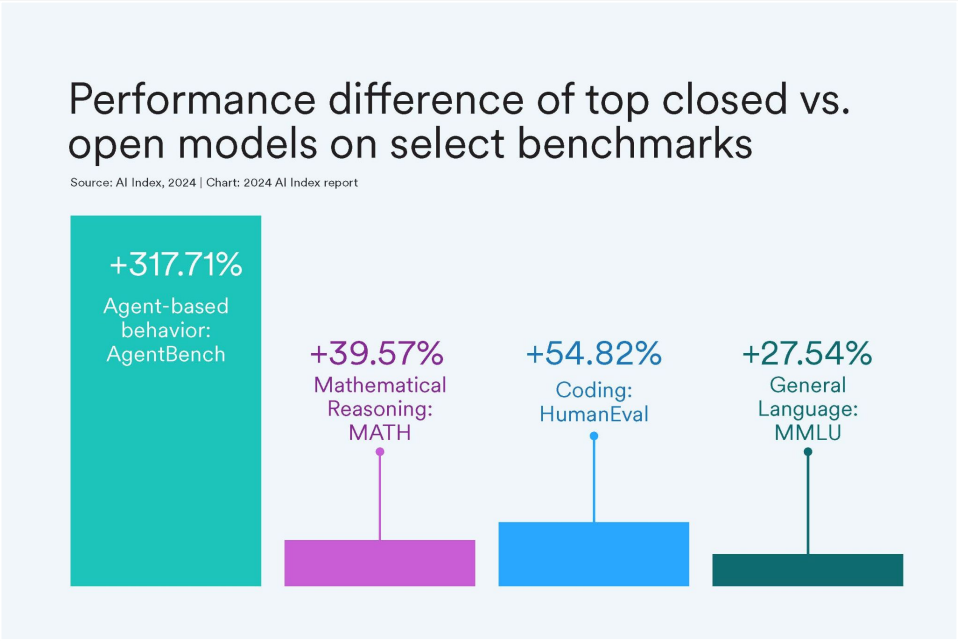
Alexa Echo

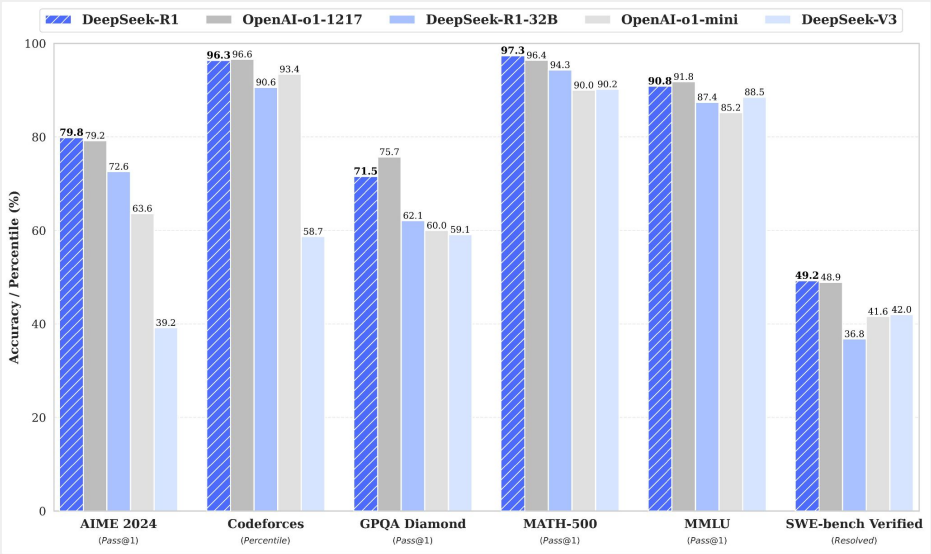




Pepper, humanoid

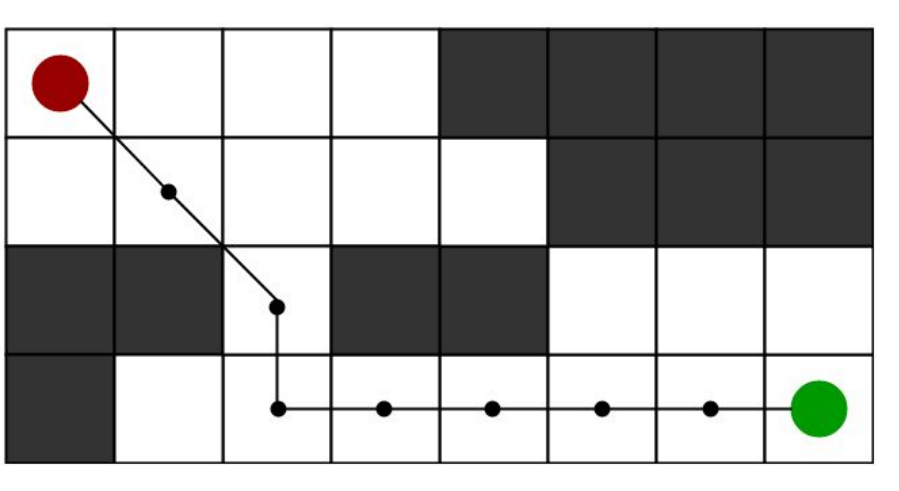
Types of AI: Open-source & Closed-source





Deepseek R1

Types of AI: Logic-based & Data-based



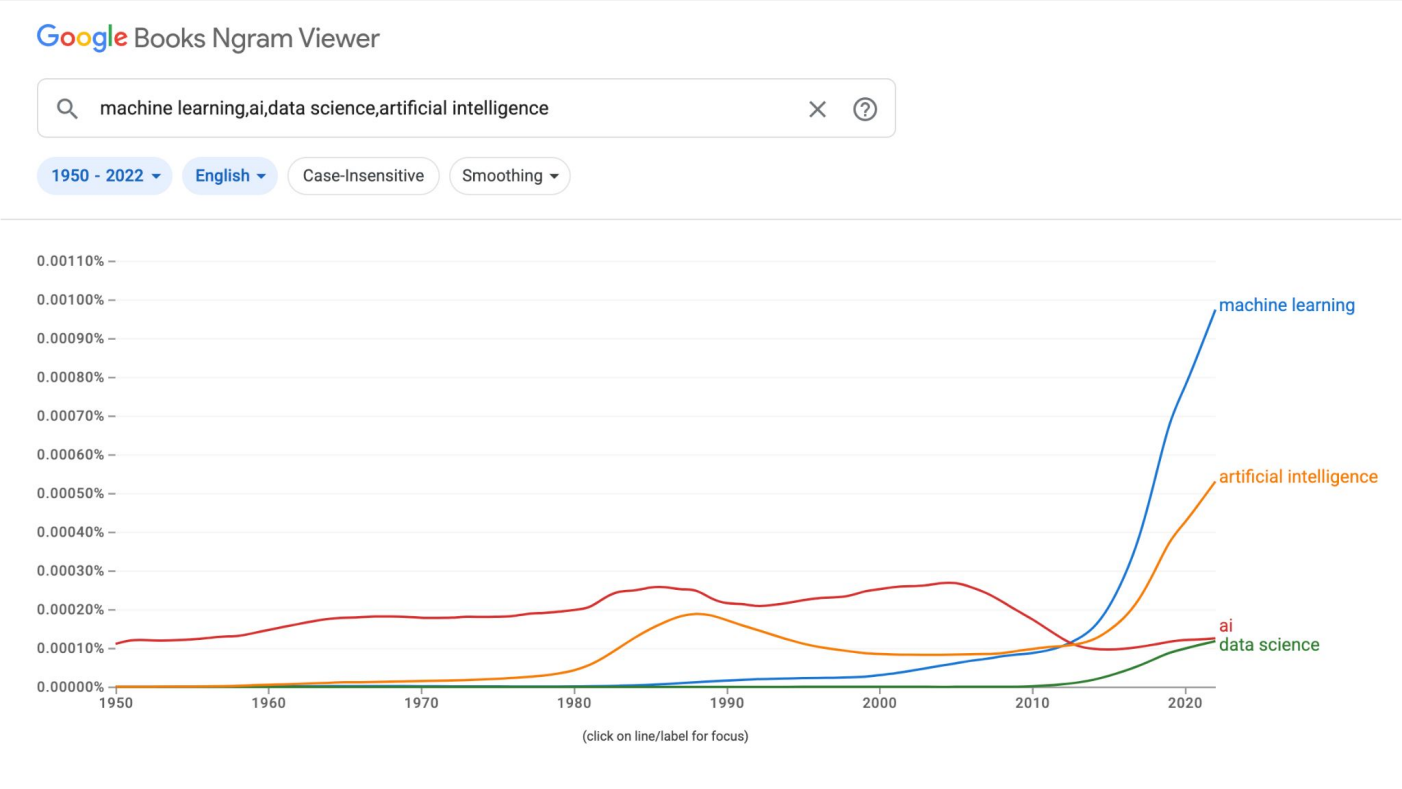
A* search

Dataset	Sampling prop.	Epochs	Disk size
CommonCrawl	67.0%	1.10	3.3 TB
C4	15.0%	1.06	783 GB
Github	4.5%	0.64	328 GB
Wikipedia	4.5%	2.45	83 GB
Books	4.5%	2.23	85 GB
ArXiv	2.5%	1.06	92 GB
StackExchange	2.0%	1.03	78 GB

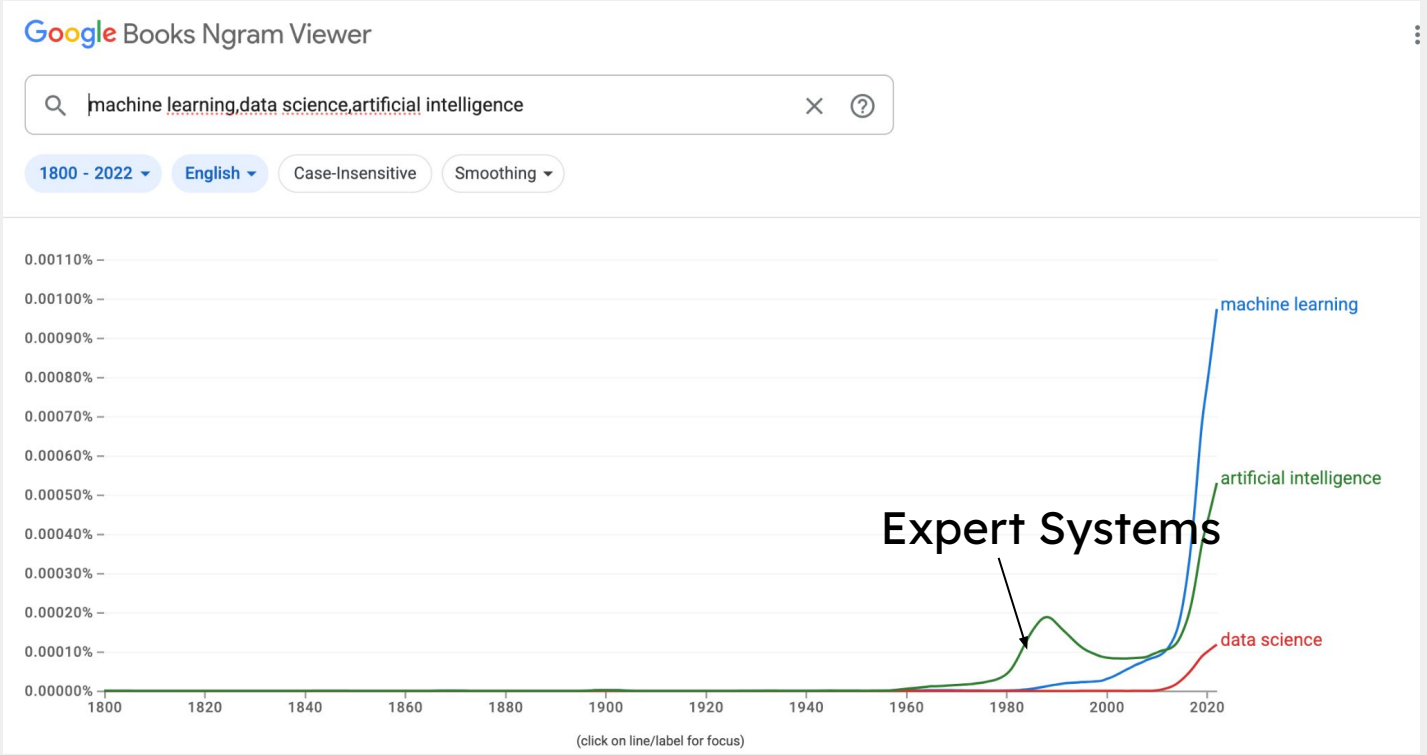
Table 1: **Pre-training data.** Data mixtures used for pre-training, for each subset we list the sampling proportion, number of epochs performed on the subset when training on 1.4T tokens, and disk size. The pre-training runs on 1T tokens have the same sampling proportion.

LlaMa Train Data

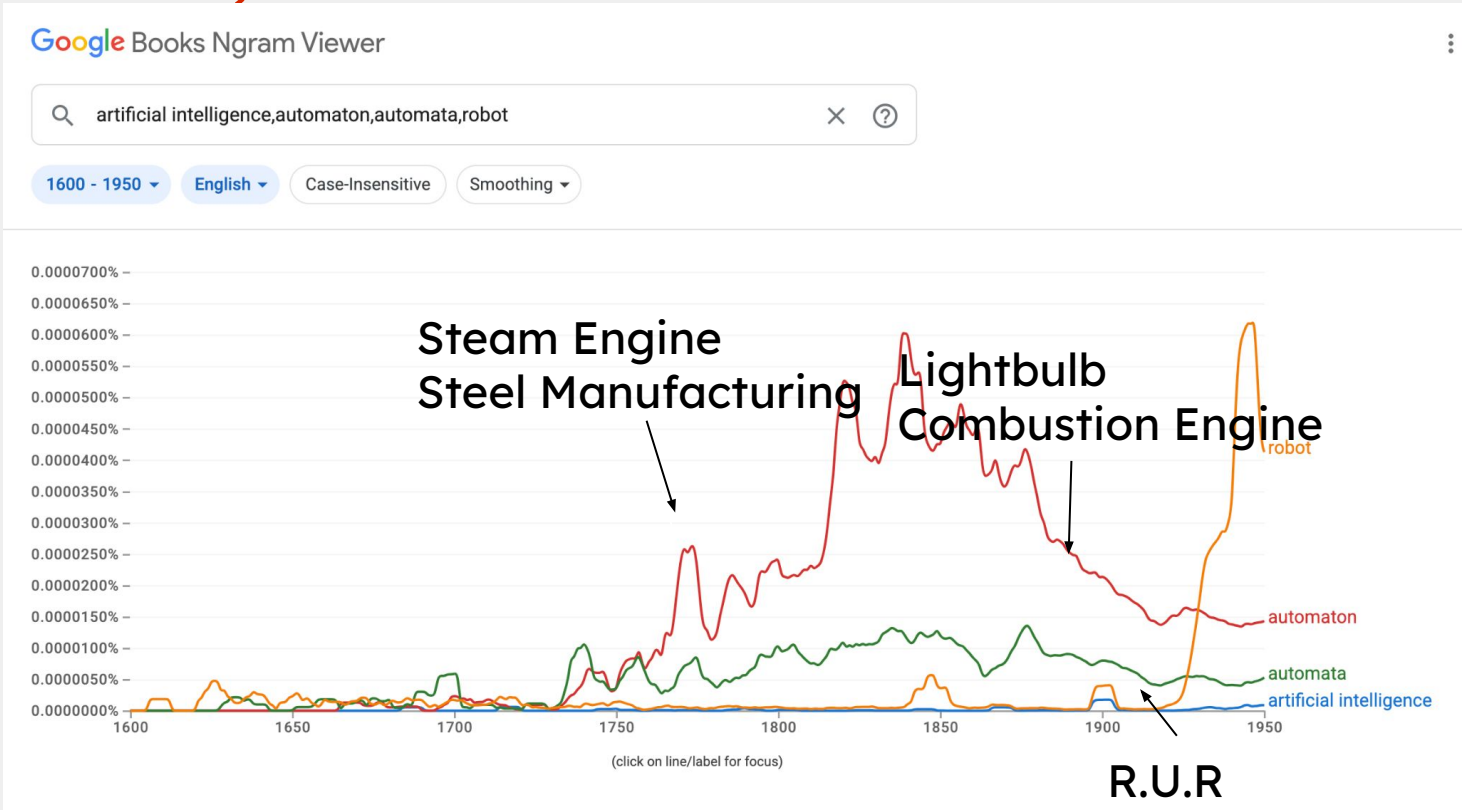
Since when, AI?



Since when, AI?



Since when, AI?



AI History

Since when, AI? + What is AI anyway?

Early Modern



Modern



Large-Scale AI?



Early Modern

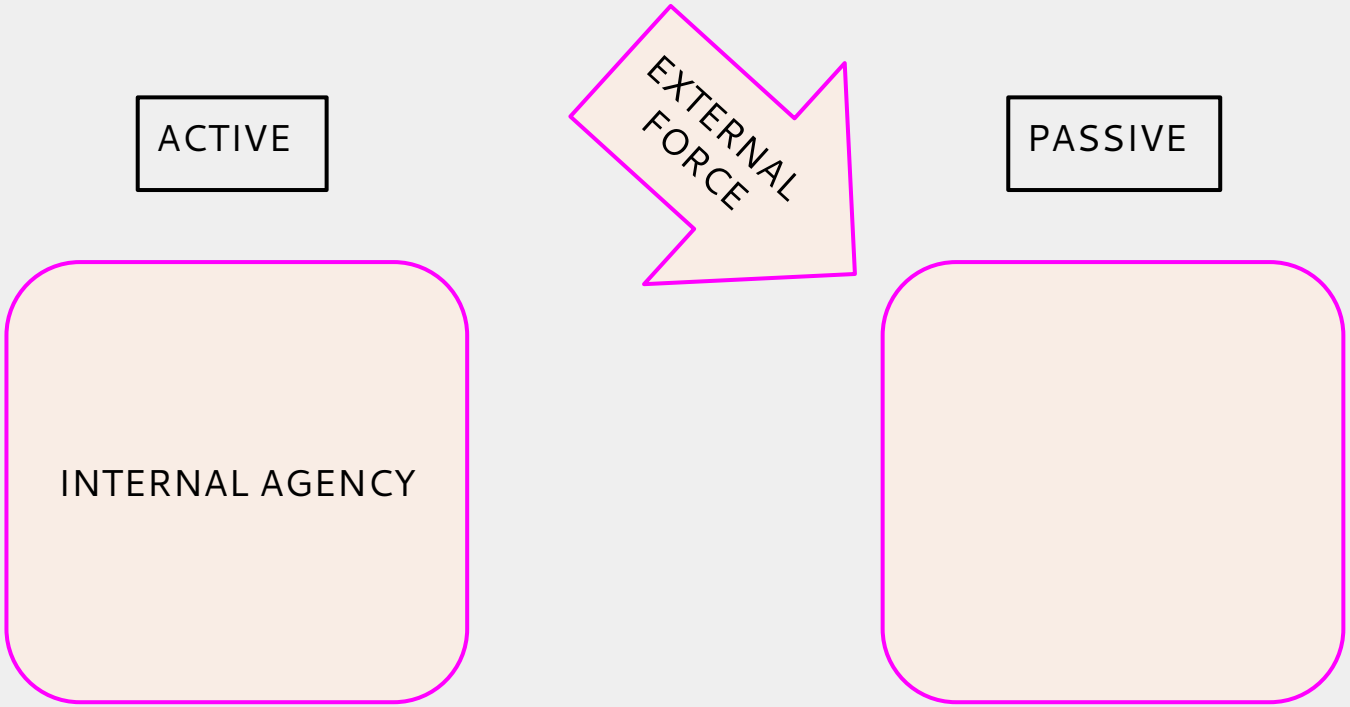
16thC Protestant Reformation & Mind-Body Separation





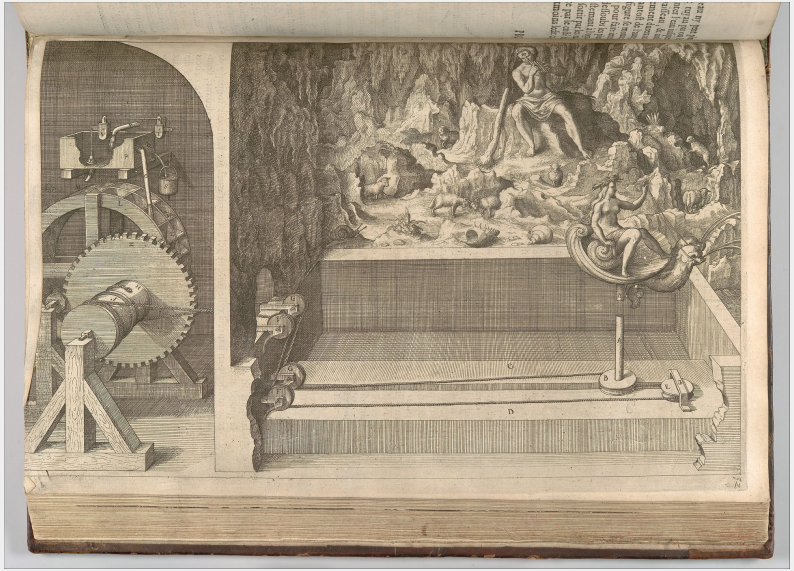
Early Modern

17thC Debates on “Agency” in Nature

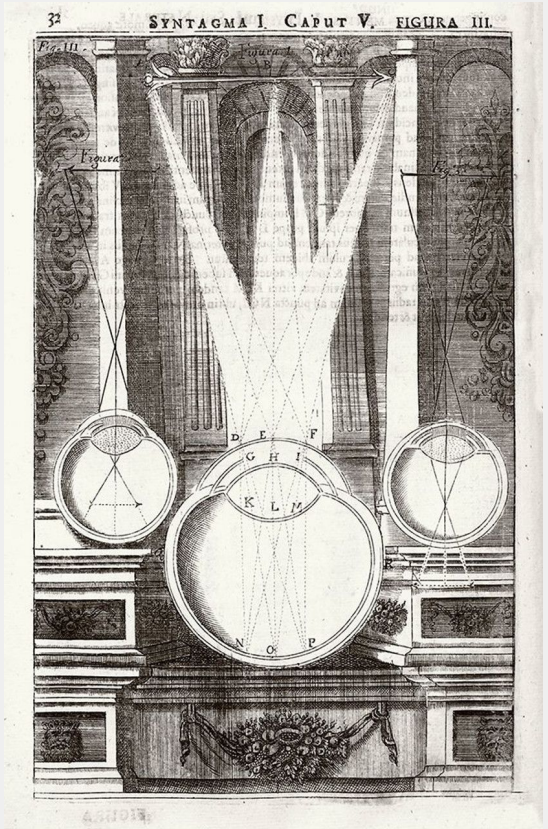


Early Modern

New Science: “Nature is God’s Artifact”



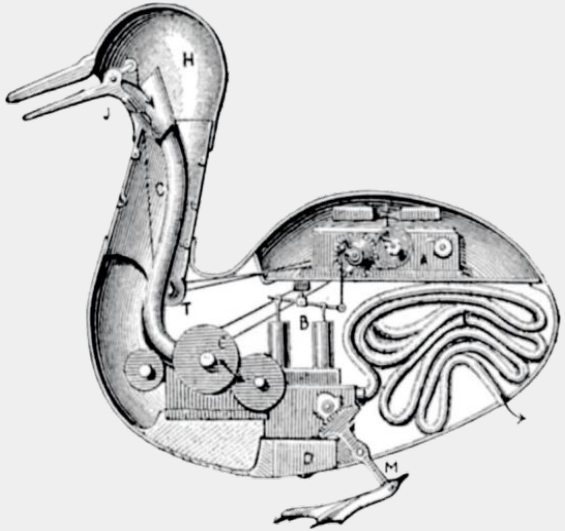
Les raisons des forces mouvantes,
Isaac De Cause. 1615

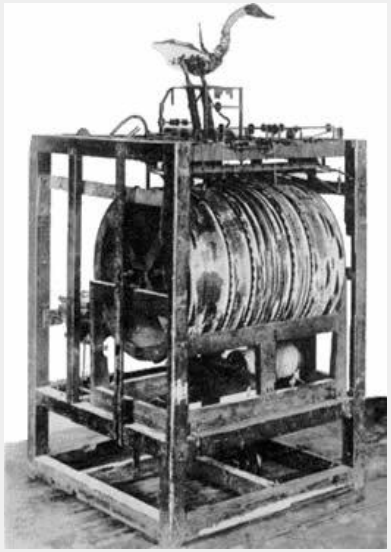


Oculus Artificialis, Johann Zahn. 1685

Early Modern

Enlightenment's first "Androids"





Canard Digérateur, Jacques Vaucanson. 1764



Mecanisme du fluteur automate, Vaucanson. 1738

Jacquard Loom 1804



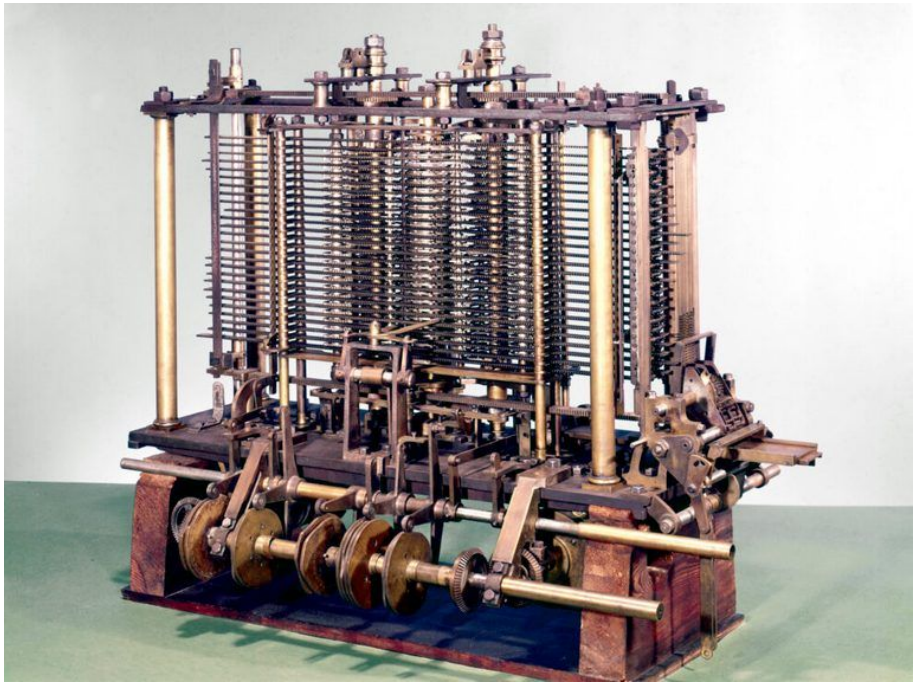


Jacquard woven on Silk. 1839





1820s Start of Computation



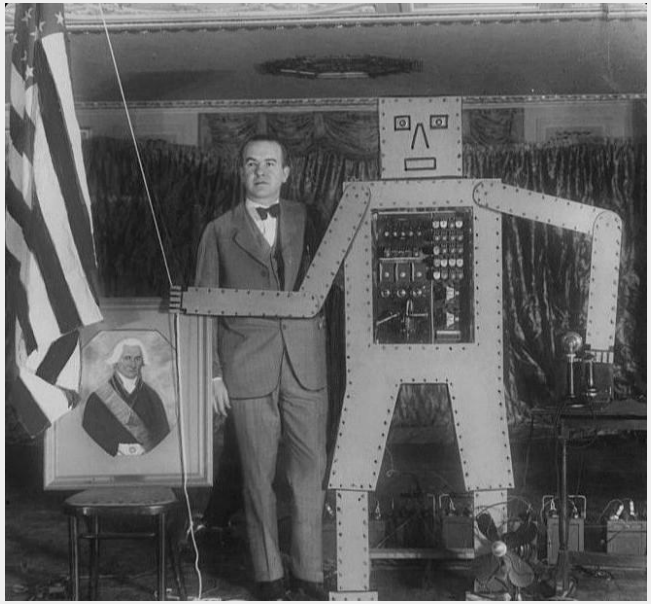
Difference Machine, Charles Babbage. 1820s



Punch Cards, Analytical Machine. Charles Babbage

Early Modern

New Science: “Nature is God’s Artifact”



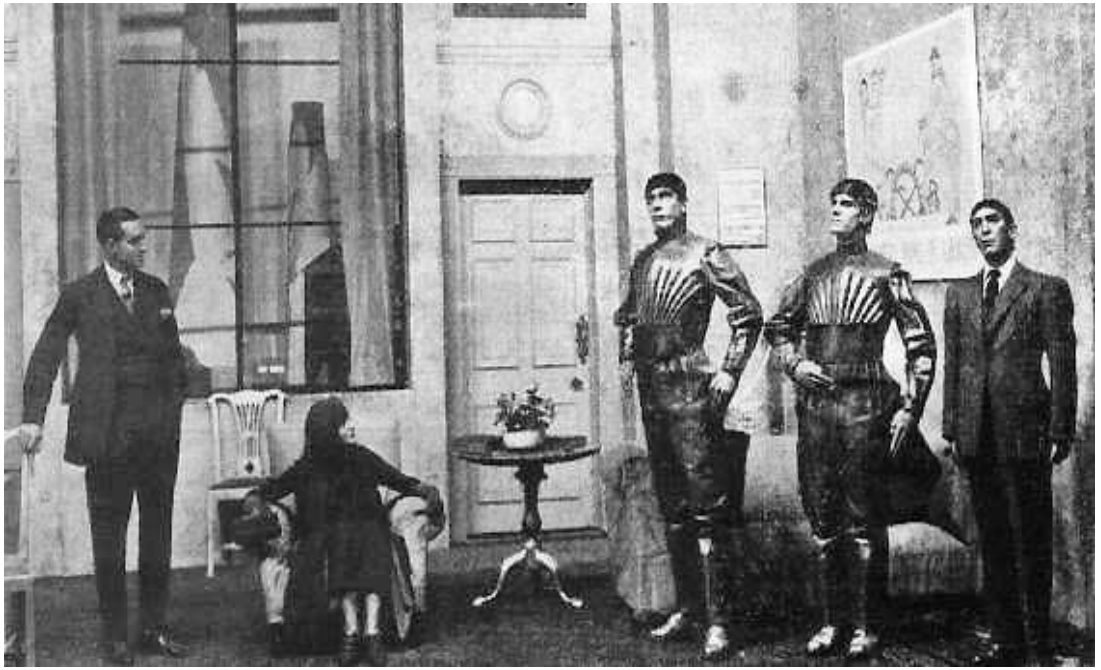
Mr. Televox. R.J. Wensley, Westinghouse (Westinghouse Electrical Company) 1926



Eric the robot at Model Engineer Exhibition at the Royal Agricultural Hall in London. 1928

“Robot” replaces “Automation” & “Android”

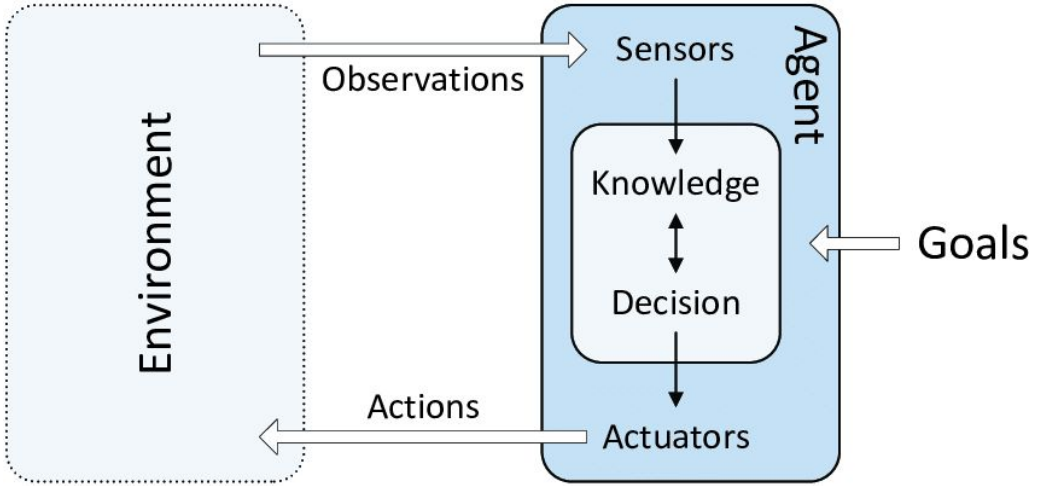




“Rossum’s Universal Robots,” Karel čapek. 1921

...to Modern

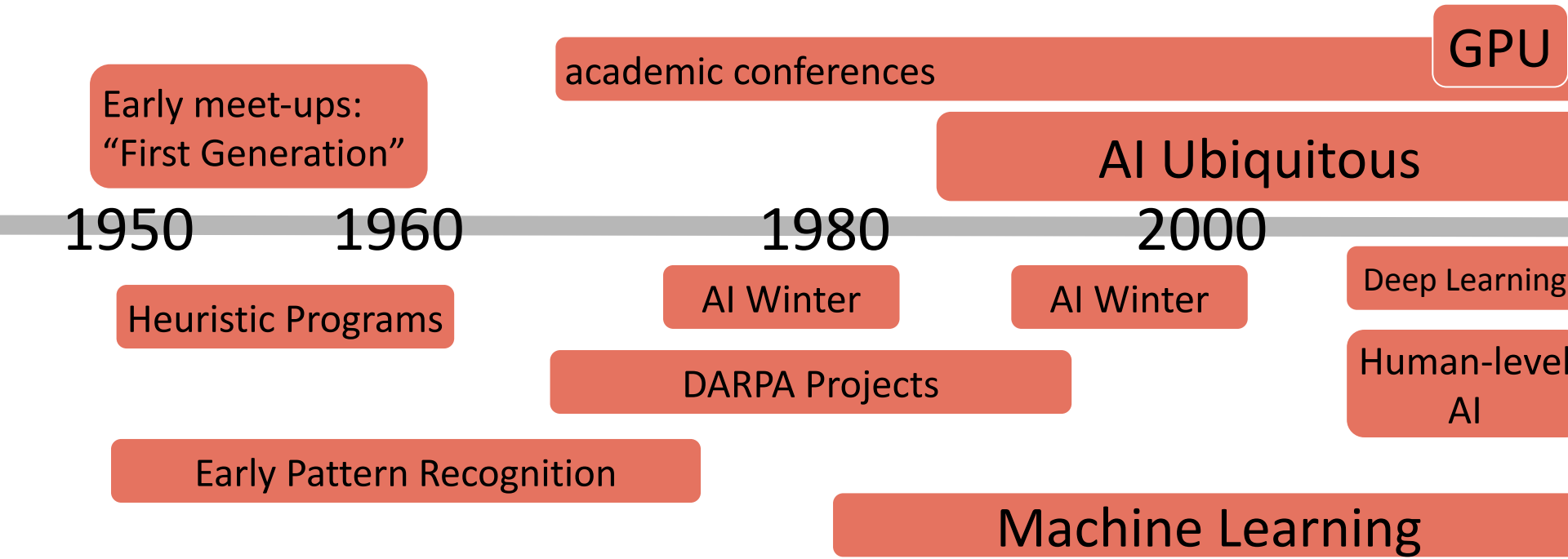
```
if status=dirty then
return clean
else if status=clean
return turn_right
else return turn_left
```



AI: A Modern Approach (Peter Norvig, Stuart Russell)

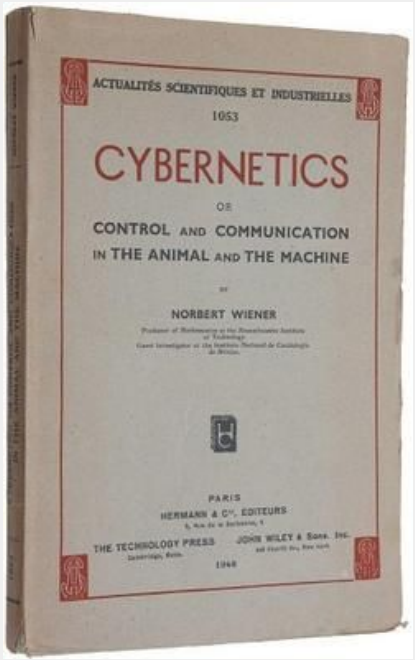
AI History

Post-war “Modern” AI

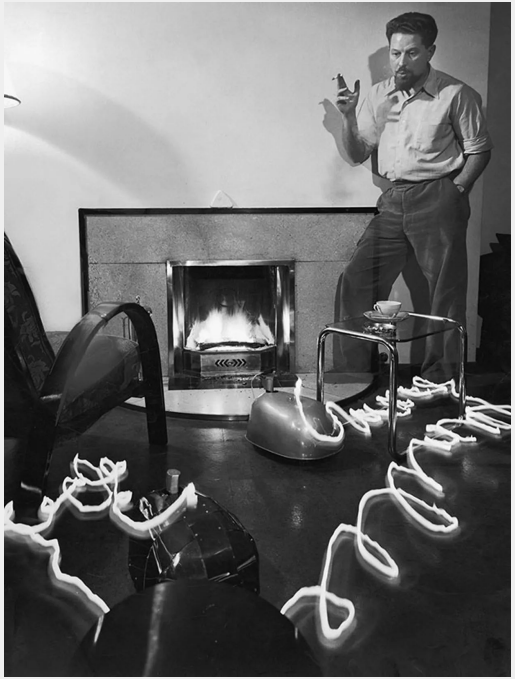


Modern

Cybernetics early & mid 20th C



Norbert Weiner. 1949



Machina speculatrix. Grey Walter. 1948

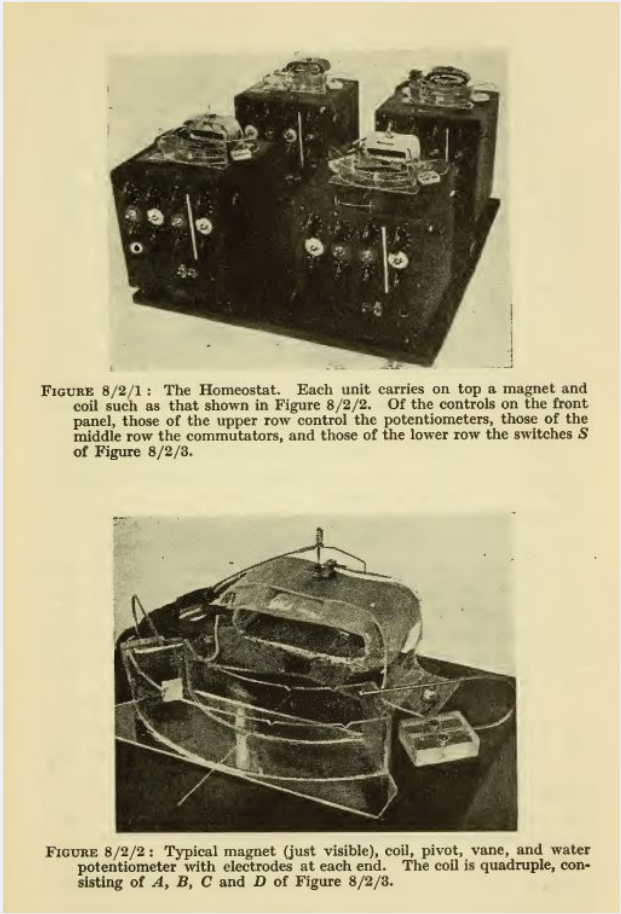


FIGURE 8/2/1 : The Homeostat. Each unit carries on top a magnet and coil such as that shown in Figure 8/2/2. Of the controls on the front panel, those of the upper row control the potentiometers, those of the middle row the commutators, and those of the lower row the switches *S* of Figure 8/2/3.

FIGURE 8/2/2 : Typical magnet (just visible), coil, pivot, vane, and water potentiometer with electrodes at each end. The coil is quadruple, consisting of *A*, *B*, *C* and *D* of Figure 8/2/3.

Homeostat with Negative Feedback. Walter Ross Ashby. 1949

Mid-20th C First digital computers
Human Mind/Intelligence ⟷ Computers

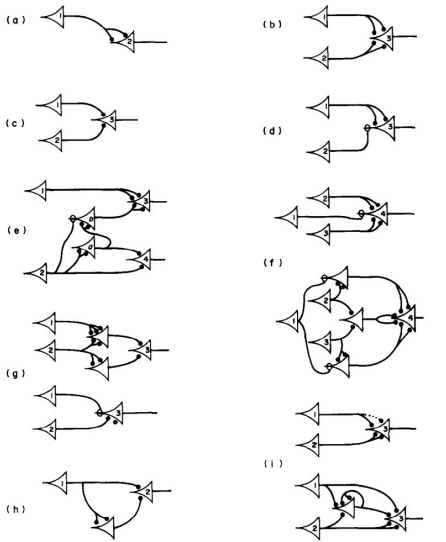
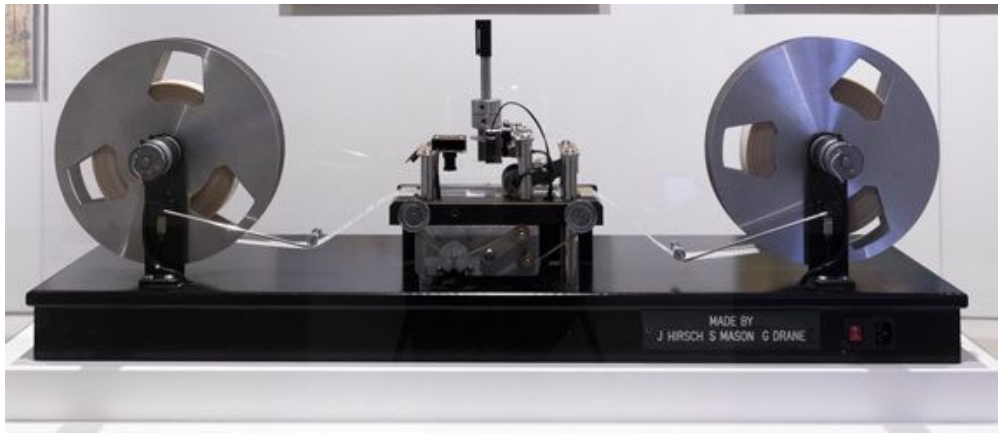


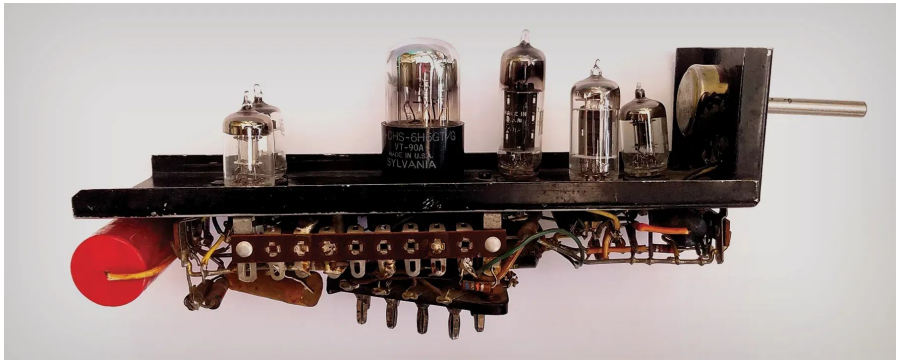
Figure 1. The neuron c_i is always marked with the numeral i upon the body of the cell, and the corresponding action is denoted by "N" with i s subscript, as in the text:

Pitts-McCulloch "Artificial Neural Network Theory." 1943

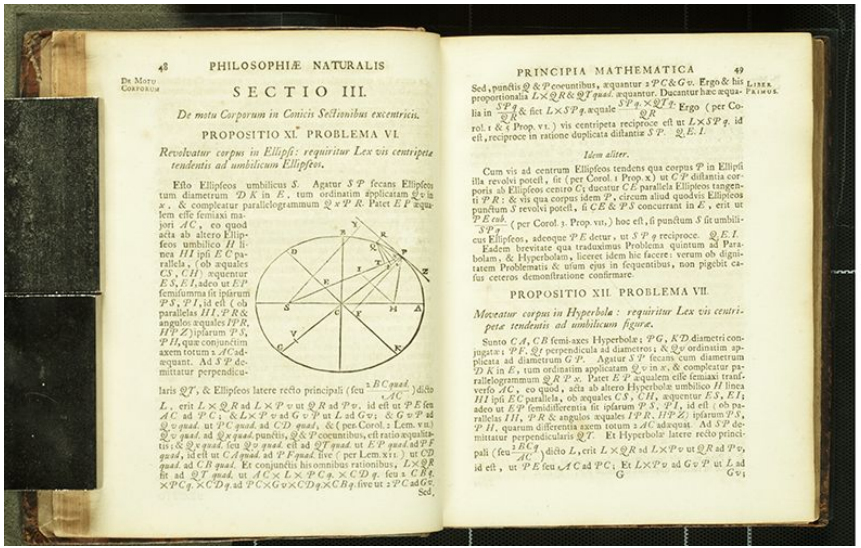


Physical model of Alan Turing's 1936 Turing Machine

First AI “programs” in the 1950s

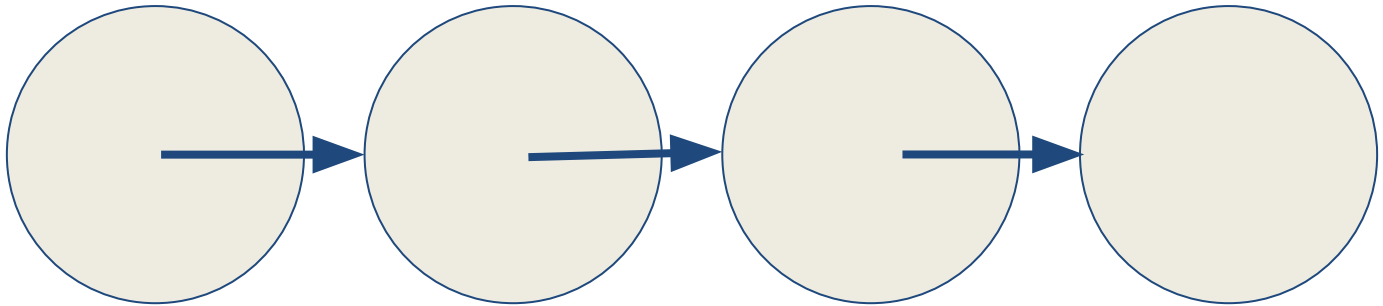


SNARC (Stochastic Neural Analog Reinforcement Calculator), Minsky-Edmonds. 1951



Logic Theorist. RAND (Newell, Simon, and Shaw). 1955

IPL (Information Processing Language) 1956 & LISP 1960 as break from FORTRAN/BASIC



1. thoughts are constantly creating, changing, and destroying
2. the links that connect symbols to other symbols in our minds can be learned

Defining "AI" and Interdisciplinary Organizations

"We propose that a 2 month, 10 man (**John McCarthy hosted; Minsky, Claude Shannon, Arthur Samuel, Ray Solomonoff, etc.**) study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hamsphire."

→ Stanford, CMU, MIT, IBM



Dartmouth Conference 1957

Cold War & Machine Translation

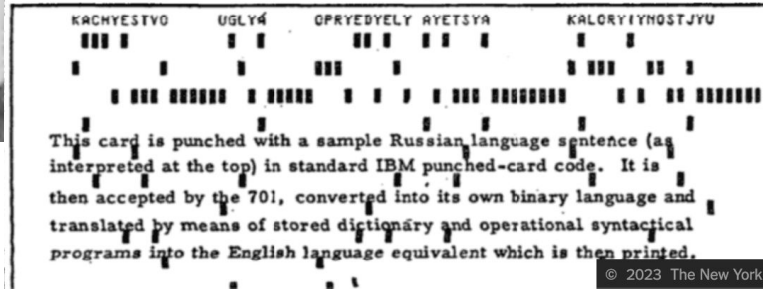


IBM 701 Electronic Data Processing Machine. 1954



The New York Tim

An electronic calculator produced by the International Business Machines Corporation wa demonstrated yesterday in a new role: translating Russian phrases into English. The de vice has a "vocabulary" of 250 Russian words and can be adapted to other languages. Mis Marilyn Polle "types" Russian phrases on I. B. M. punch cards that are fed into the machine



This card is punched with a sample Russian language sentence (as interpreted at the top) in standard IBM punched-card code. It is then accepted by the 701, converted into its own binary language and translated by means of stored dictionary and operational syntactical programs into the English language equivalent which is then printed.

Georgetown–IBM Russian Translation Demo. 1954

© 2023 The New York

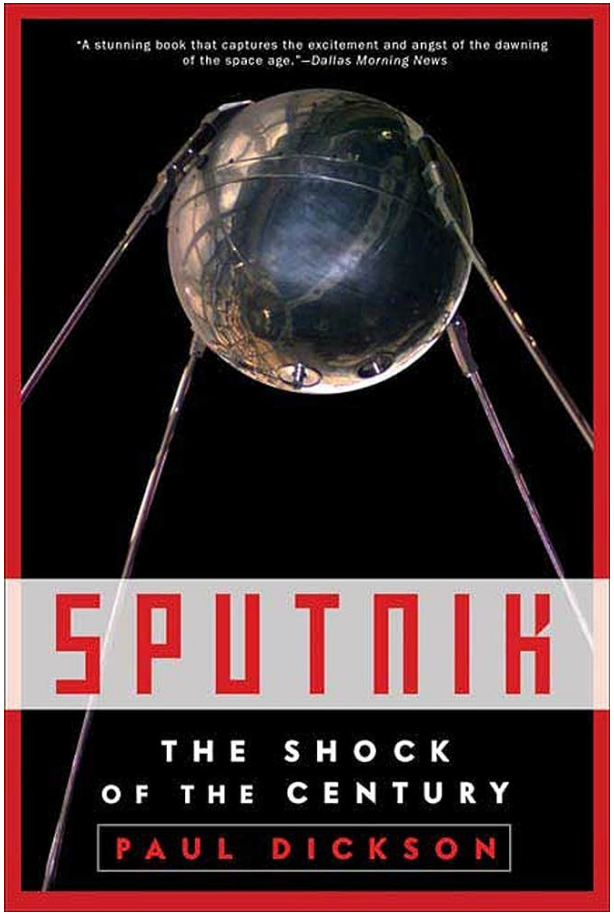
Ruský originál (a angl. transkripce)	Anglický překlad
Качество угля определяется калорийностью Kachyestvo uglya opryedyelyayetsya kaloryiynostjyu	The quality of coal is determined by calory content
Крахмал вырабатывается механическим путем из картофеля Kraxmal virabativayetsya myexanyichyeskyim putyem yiz kartofyelya	Starch is produced by mechanical methods from potatoes
Железо добывается из руды химическим процессом Zhyelyezo dobivayetsya yiz rudi xyimyicheskyim protsyessom	Iron is obtained from ore by chemical process
Мы передаем мысли посредством речи Mi pyeryedayem mislyi posryedstvom ryechi	We transmit thoughts by means of speech
Величина угла определяется отношением длины дуги к радиусу Vyelyichyina ugla opryedyelyayetsya otnoshyenyiyem dlyini dugi k radyiusu	Magnitude of angle is determined by the relation of length of arc to radius
Международное понимание является важным фактором в решении политических вопросов Myezhdunarodnoye ponyimanyiye yavlyayetsya vazhnim faktorom v ryeshyenyiyi polyityichyeskyix voprosov	International understanding constitutes an important factor in decision of political questions
Владимир является на работу поздно утром Vladyimir yavlyayetsya na rabotu pozdno utrom	Vladimir appears for work late in the morning

250 Russian words, six syntax rules → 60 Russian sentences to English

1957 Sputnik Launch, 1958 ARPA

ARPA

ADVANCED RESEARCH PROJECTS AGENCY



1961 Licklider heads IPTO (Information Processing Techniques Office)

1. Time: Batch-processing was too slow → need time sharing for real-time, interactive computing (especially in command-and-control)

2. Language: Assembly → programming languages more similar to human beings (IPL/LISP?); Computers and humans should be able to interact in ordinary speech

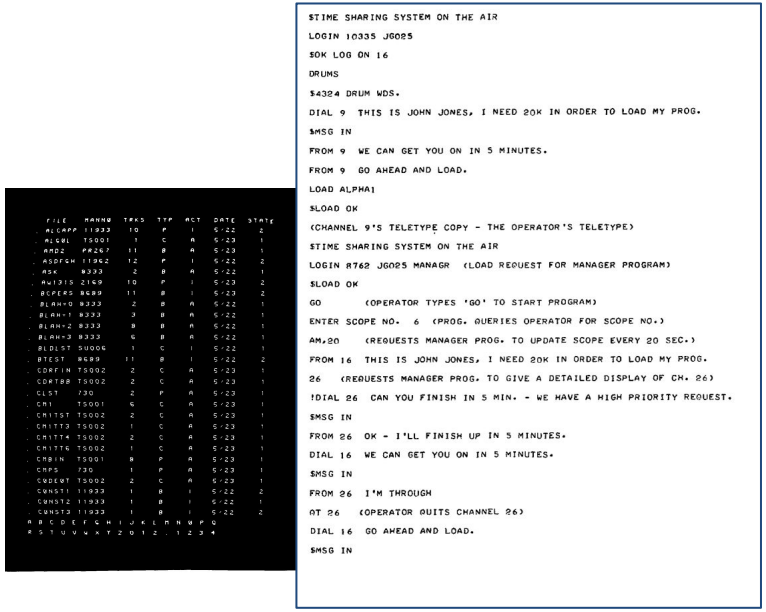
"The hope is that, in not too many years, human brains and computing machines will be coupled together very tightly, and that the resulting partnership will think as no human brain has ever thought and process data in a new way not approached by the information-handling machines we know today."

JCR Licklider, Man-Computer Symbiosis, 1960

JCR Licklider



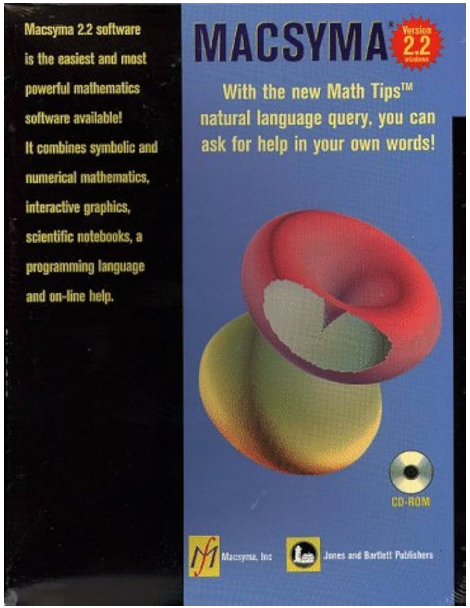
1961 Licklider'sIPTO
(Information Processing Techniques Office)



```
 FILE   MANNO  TRKS  TTP  ACT  DATE  STATE
 ALCAPP 11933   10    P    I   5-22    2
 ALGBL  TS001    1    C    A   5-23    1
 AMD2   P8267   11    B    A   5-23    1
 ASDFGH 11962   12    P    I   5-22    2
 ASK     8333    2    B    A   5-22    1
 AW13IS 2169    10    P    I   5-23    2
 BEPERS 8689    11    B    I   5-23    2
 BLAH-0 8333     2    B    A   5-22    1
 BLAH-1 8333     3    B    A   5-22    1
 BLAH-2 8333     8    B    A   5-22    1
 BLAH-3 8333     6    B    A   5-22    1
 BLDLST SU006    1    C    I   5-22    1
 BTEST  8689    11    B    I   5-22    2
 CORFIN TS002    2    C    A   5-23    1
 CORTBB TS002    2    C    A   5-23    1
 CLST    730     2    P    A   5-23    1
 CM1    TS001    6    C    A   5-23    1
 CM1TST TS002    2    C    A   5-23    1
 CM1TT3 TS002    1    C    A   5-23    1
 CM1TT4 TS002    2    C    A   5-23    1
 CM1TTG TS002    1    C    A   5-23    1
 CMBIN  TS001    8    P    A   5-23    1
 CMPS    730     1    P    A   5-23    1
 CWDERT TS002    2    C    A   5-23    1
 CWNST1 11933    1    B    I   5-22    2
 CWNST2 11933    1    B    I   5-22    1
 CWNST3 11933    1    B    I   5-22    2
A B C D E F G H I J K L M N O P Q
R S T U V W X Y Z 0 1 2 . 1 2 3 4
```

```
$TIME SHARING SYSTEM ON THE AIR
LOGIN 10335 JG025
$OK LOG ON 16
DRUMS
$4324 DRUM WDS.
DIAL 9  THIS IS JOHN JONES, I NEED 20K IN ORDER TO LOAD MY PROG.
$MSG IN
FROM 9  WE CAN GET YOU ON IN 5 MINUTES.
FROM 9  GO AHEAD AND LOAD.
LOAD ALPHA1
$LOAD OK
(CHANNEL 9'S TELETYPE COPY - THE OPERATOR'S TELETYPE)
$TIME SHARING SYSTEM ON THE AIR
LOGIN R762 JG025 MANAGR  (LOAD REQUEST FOR MANAGER PROGRAM)
$LOAD OK
GO       (OPERATOR TYPES 'GO' TO START PROGRAM)
ENTER SCOPE NO.  6  (PROG. QUERIES OPERATOR FOR SCOPE NO.)
AM,20    (REQUESTS MANAGER PROG. TO UPDATE SCOPE EVERY 20 SEC.)
FROM 16  THIS IS JOHN JONES, I NEED 20K IN ORDER TO LOAD MY PROG.
26   (REQUESTS MANAGER PROG. TO GIVE A DETAILED DISPLAY OF CH. 26)
!DIAL 26  CAN YOU FINISH IN 5 MIN. - WE HAVE A HIGH PRIORITY REQUEST.
$MSG IN
FROM 26  OK - I'LL FINISH UP IN 5 MINUTES.
DIAL 16  WE CAN GET YOU ON IN 5 MINUTES.
$MSG IN
FROM 26  I'M THROUGH
OT 26   (OPERATOR QUITS CHANNEL 26)
DIAL 16  GO AHEAD AND LOAD.
$MSG IN
```

TSS: Time-sharing with Graphical User Interface → becomes icon-based GUI for windows/Apple



MIT's MAC (Man and Computer, Machine-aided Cognition, Multi-Access Computing): ARPA spent $25 million on Project MAC between 1963-1970

ARPA gives birth to AI labs across U.S.

- @MIT: Marvin Minsky and students
- @Stanford: John McCarthy started the Stanford AI Lab with unconditional ARPA money 1964
- ARPA supported AI at Rand, Carnegie-Mellon, SRI, SDC ,BBN, and seven other “centers of excellence”
- By the mid-1970s, IPTO’s budget accounted for 40-50% of applied research in math and computers
- 80-90% of funding for major AI groups (MIT, Stanford, SRI, CMU) came from ARPA/DARPA

1960s Memory-Bounded Methods

We shall illustrate the way the *advice taker* is supposed to act by means of an example. Assume that I am seated at my desk at home and I wish to go to the airport. My car is at my home also. The solution of the problem is to walk to the car and drive the car to the airport. First, we shall give a formal statement of the premises the *advice taker* uses to draw the conclusions. Then we shall discuss the heuristics which cause the *advice taker* to assemble these premises from the totality of facts it has available. The premises come in groups, and we shall explain the interpretation of each group.

1. First, we have a predicate "*at*". "$at(x, y)$" is a formalization of "*xisaty*". Under this heading we have the premises

$$at(I, desk) \quad (1)$$

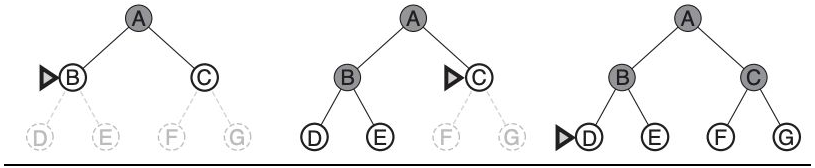
$$at(desk, home) \quad (2)$$

$$at(car, home) \quad (3)$$

$$at(home, county) \quad (4)$$

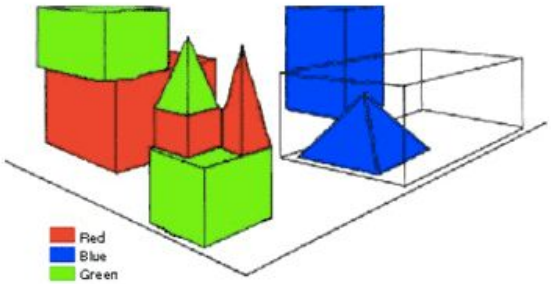
$$at(airport, county) \quad (5)$$

"advice taker," John McCarthy. 1968



Breadth-First Search. Moore 1958

Microworlds 1970s



Person: Pick up a big red block.

Computer: OK.

Person: Grasp the pyramid.

Computer: I don't understand which pyramid you mean.

SHRDLU. Terry Winograd. 1972

```
Person: PICK UP A BIG RED BLOCK.
Computer: OK. (does it)
Person: GRASP THE PYRAMID.
Computer: I DON'T UNDERSTAND WHICH PYRAMID YOU MEAN.
Person: FIND A BLOCK WHICH IS TALLER THAN THE ONE YOU ARE HOLDING AND PUT IT INTO THE BOX.
Computer: BY "IT", I ASSUME YOU MEAN THE BLOCK WHICH IS TALLER THAN THE ONE I AM HOLDING.
Computer: OK. (does it)
Person: WHAT DOES THE BOX CONTAIN?
Computer: THE BLUE PYRAMID AND THE BLUE BLOCK.
Person: WHAT IS THE PYRAMID SUPPORTED BY?
Computer: THE BOX.
Person: HOW MANY BLOCKS ARE NOT IN THE BOX?
Computer: FOUR OF THEM.
Person: IS AT LEAST ONE OF THEM NARROWER THAN THE ONE WHICH I TOLD YOU TO PICK UP?
Computer: YES, THE RED CUBE.
Person: IS IT SUPPORTED?
Computer: YES, BY THE TABLE.
Person: CAN THE TABLE PICK UP BLOCKS?
Computer: NO.
Person: CAN A PYRAMID BE SUPPORTED BY A BLOCK?
Computer: YES.
Person: CAN A PYRAMID SUPPORT A PYRAMID?
Computer: I DON'T KNOW.
Person: STACK UP TWO PYRAMIDS.
Computer: I CAN'T.
Person: WILL YOU PLEASE STACK UP BOTH OF THE RED BLOCKS AND EITHER A GREEN CUBE OR A PYRAMID?
Computer: OK. (does it)
Person: WHICH CUBE IS SITTING ON THE TABLE?
Computer: THE LARGE GREEN ONE WHICH SUPPORTS THE RED PYRAMID.
Person: IS THERE A LARGE BLOCK BEHIND A PYRAMID?
Computer: YES, THREE OF THEM: A LARGE RED ONE, A LARGE GREEN CUBE, AND THE BLUE ONE.
Person: PUT A SMALL ONE ONTO THE GREEN CUBE WHICH SUPPORTS A PYRAMID.
Computer: OK. (does it)
Person: PUT THE LITTLEST PYRAMID ON TOP OF IT.
Computer: OK. (does it)
Person: DOES THE SHORTEST THING THE TALLEST PYRAMID'S SUPPORT SUPPORTS SUPPORT ANYTHING GREEN?
Computer: YES, THE GREEN PYRAMID.
Person: WHAT COLOR IS IT?
Computer: BY "IT", I ASSUME YOU MEAN THE SHORTEST THING THE TALLEST PYRAMID'S SUPPORT SUPPORTS.
Computer: RED
Person: HOW MANY THINGS ARE ON TOP OF GREEN CUBES?
Computer: I'M NOT SURE WHAT YOU MEAN BY "ON TOP OF "IN THE PHRASE "ON TOP OF GREEN CUBES " .
 DO YOU MEAN:
```

ARPA & the Vietnam War (Kennedy's project AGILE)



ARPA's first drones ca. 1968



Sensors dropped on Ho Chi Minh trail

ARPA rebrands to DARPA 1972

Greater accountability → more paperwork and less freedom, constrained autonomy

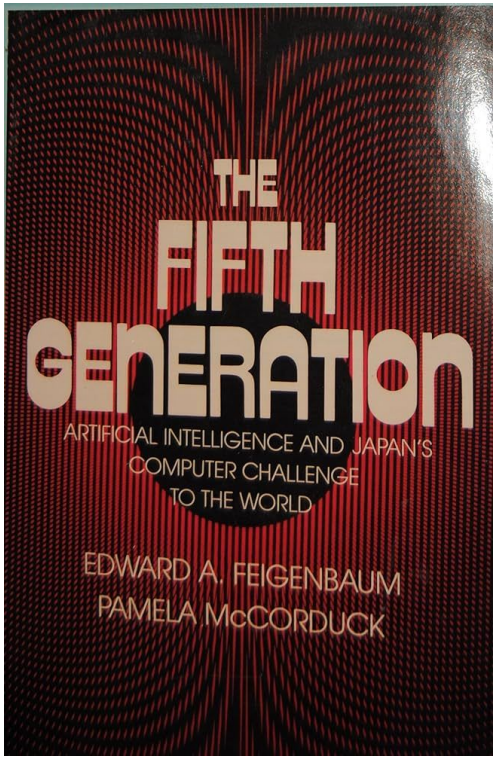
Only Applications-Focused Projects



Reagan's Strategic Defense Initiative and Strategic Computing Initiative (1983 - 1993)







- 10 Year Projects: Explicit relation to battlefield technology
- By 1985, DARPA (under IPTO guidance) had spent $100 million for 92 projects underway at 60 institutions (half in industry, half in universities and gov)



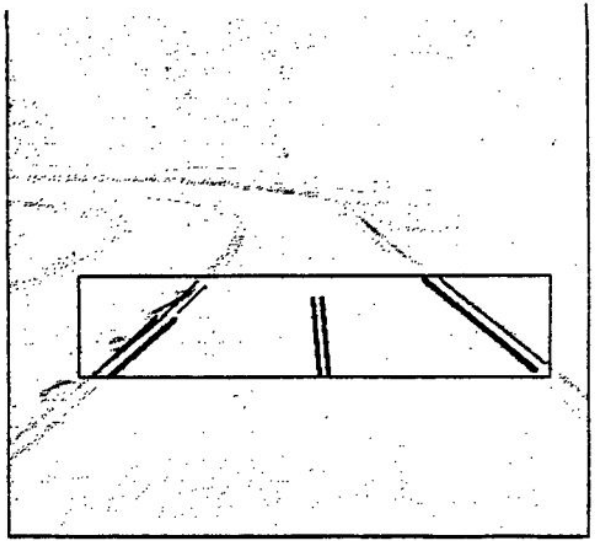
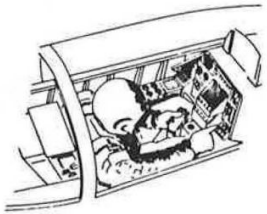


Figure 2: Road edge detection. Here a road image is shown after processing to enhance intensity changes. The vision program selects a window in which to search for road edges. Candidate left and right road edges are lines fit to the raw edge data, shown here as black lines. Heavy black lines indicate the left and right road edges selected by the program. The computed road center line is shown as a double line

Autonomous Land Vehicle Project (1984–1988) at CMU

Figure 3. Pilot's Associate Program Overview.



FUNCTIONAL AREAS

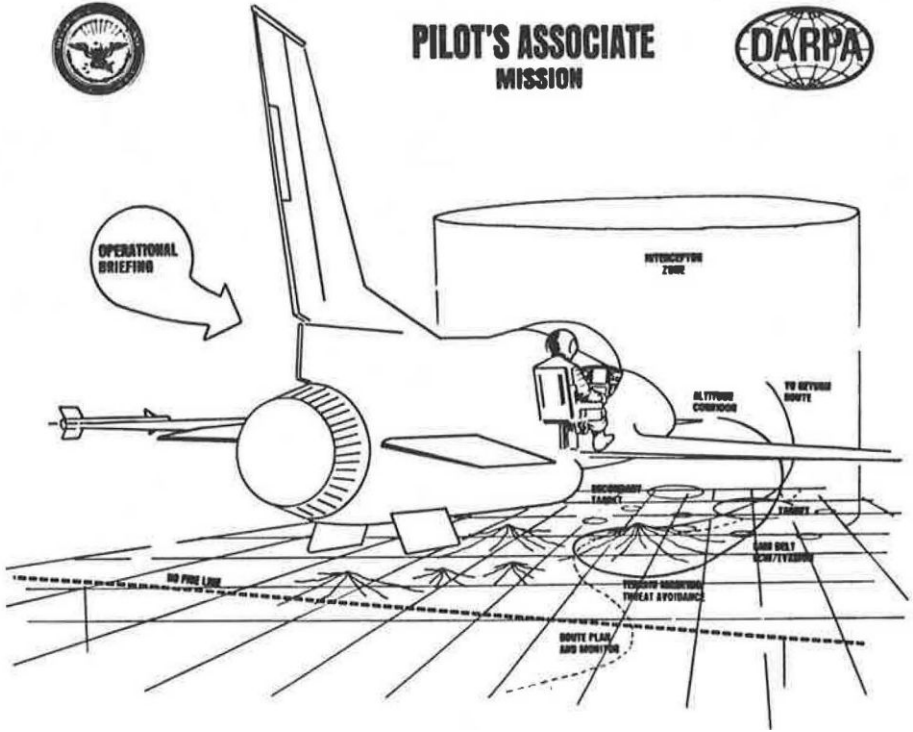
- SITUATION ASSESSMENT
- SYSTEM STATUS
- MISSION PLANNING
- TACTICS ADVISORY

DEVELOPMENT ENVIRONMENT

- COMPUTER LABORATORY -- PART TASK SIMULATORS
- FULL MISSION SIMULATORS

6643/88W/GW28/1

Figure 6. Pilot's Associate Mission.



Intelligent fighter assistant for the Air Force (Pilot's Associate)

Problem-Solving → Knowledge-Based AI 1980s

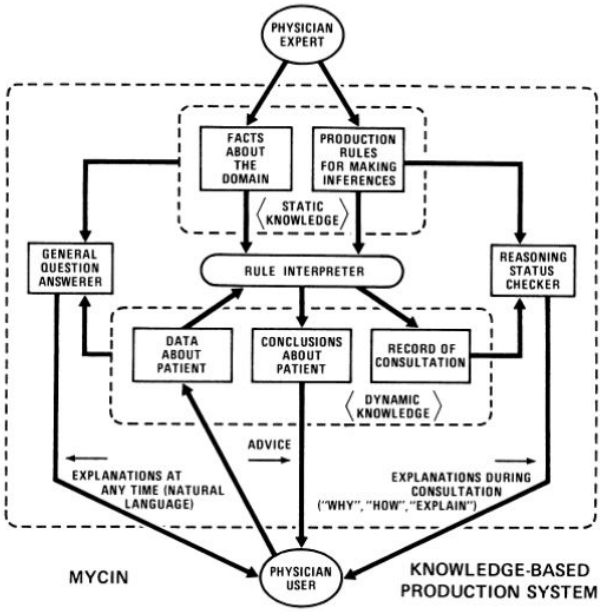
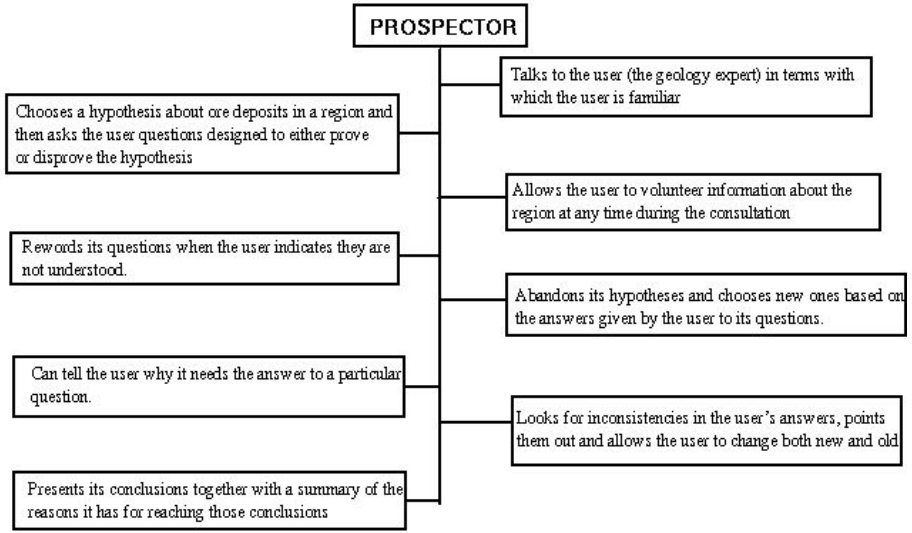


Figure 2 - Diagram summarizing the organization and flow of information within MYCIN. The correlation between this design and the human consultation process depicted in Fig. 1 is discussed in the text. (Figure reproduced from reference 10).

MYCIN (1970s) was accurate in about 69% of cases



PROSPECTOR (1983) helped a company discover molybdenum deposits in Washington.

1980s Start-Ups and Private Financing

Product Information

Software

- **KEE Version 3.0 for workstations**
- **KEE Version 3.0 for 80386-based IBM-compatible PCs**
- **KEEConnection**
- **SimKit**
- **Run-Time KEE**

KEE Versions 1.0 & 2.0 and Intelliscope are no longer sold.

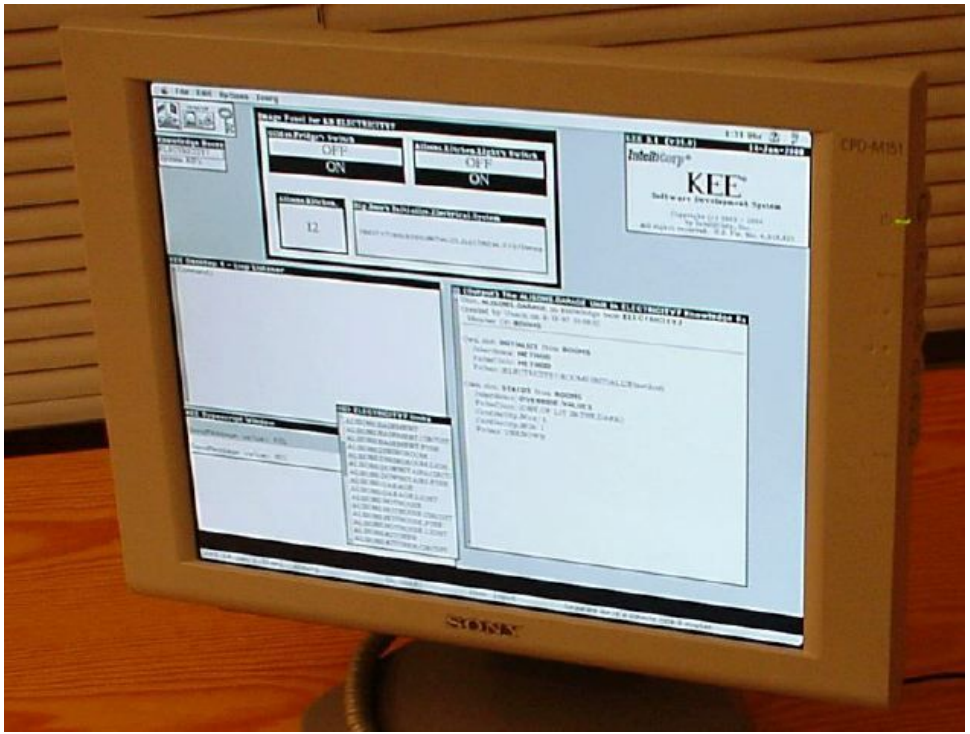
Pricing

KEE Version 3.0 (workstations)	**$46,500** academic price **$9,250**
KEE Version 3.0 (386 PCs)	**$9,900**
KEEConnection (varies with hardware)	**$15,500** academic price **$5,000**
SimKit	**$21,000** academic price **$4,125**
Run-Time KEE (single-user workstation)	**$5,000**

Pricing varies, depending on number of licenses purchased, and includes 3-9 days training for 2 persons and one year's support. Support services include standard consultants, senior scientist consultants and/or apprenticeship training (4 weeks).

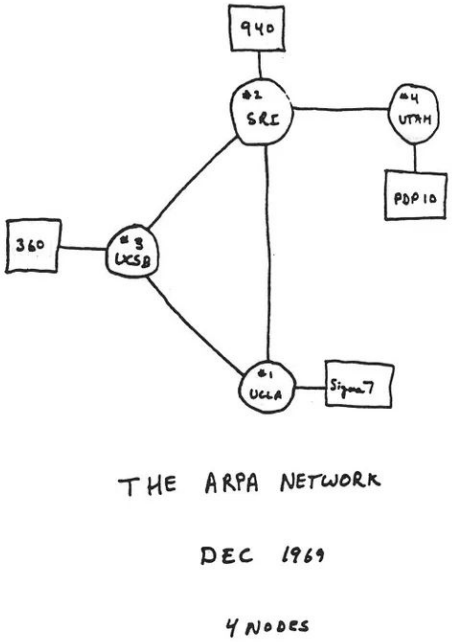
Hardware

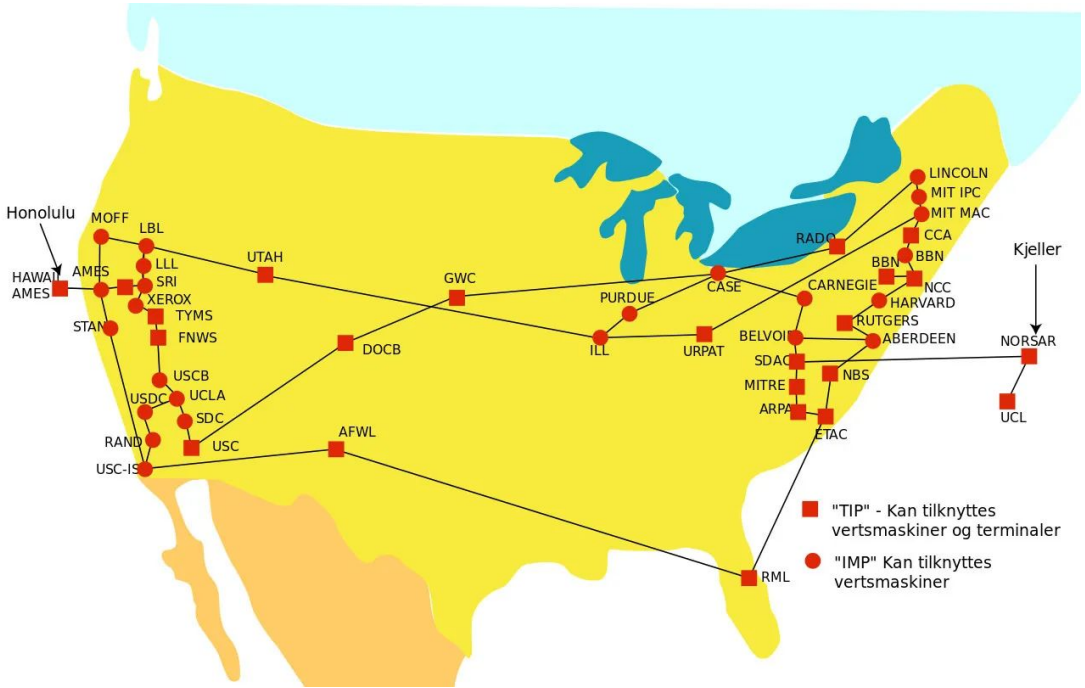
all 80386-based IBM PC-compatible computers
IBM PC RT
Symbolics 36XX series
Xerox 1100 series
Sun 3 workstations
Apollo Domain workstations
TI Explorer I and II
DEC VAX series
MicroVAX II workstations



Intellicorp's Knowledge Engineering Environment (KEE) 1983

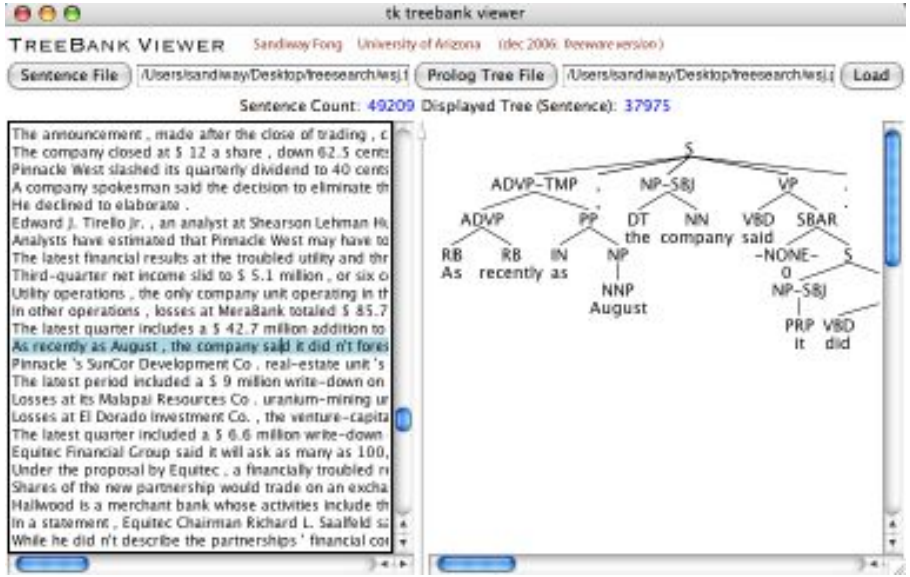
Meanwhile ARPANET 1966 → Web 1.0

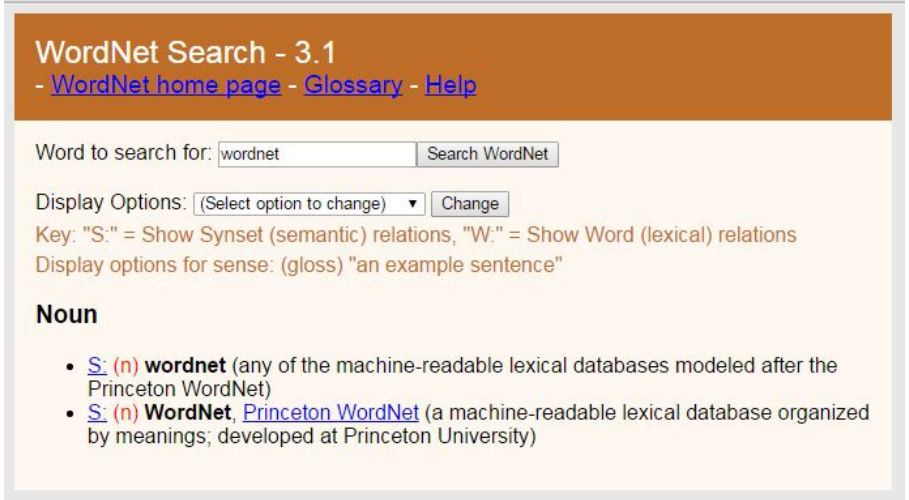




ARPANET 1970s

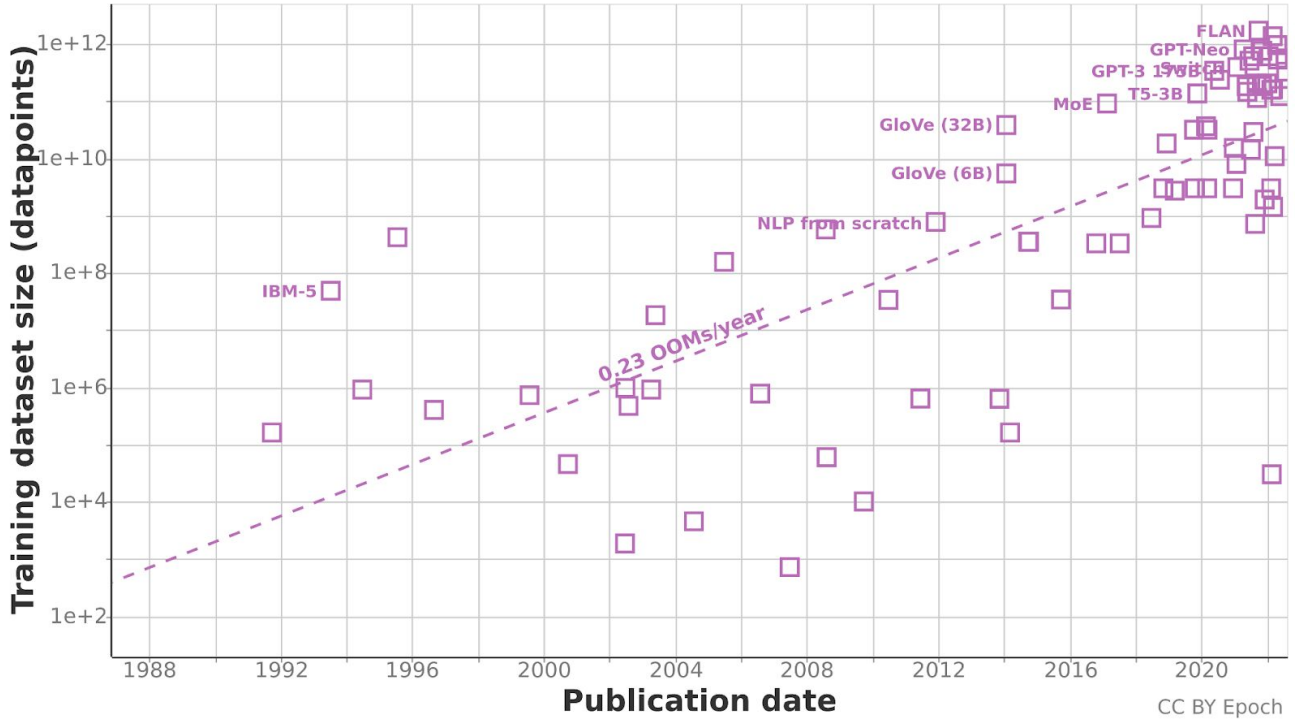
90s - 2000s Online Datasets



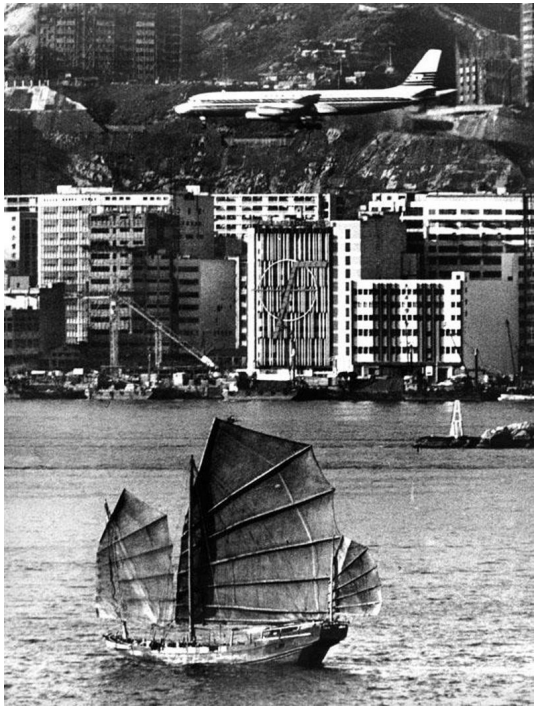


Penn Treebank (1993), WordNet (2001), British National Corpus (2001) ... by the early 2000s, the World Wide Web contained several trillion words.

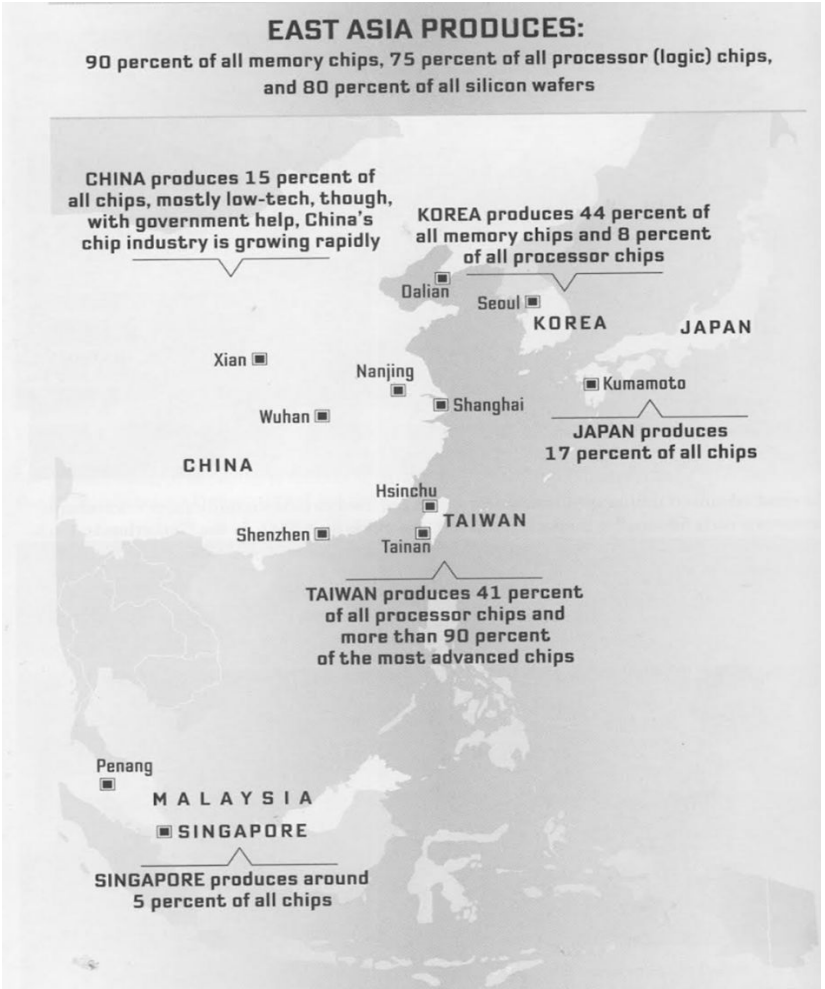
...to Large scale AI era(?)



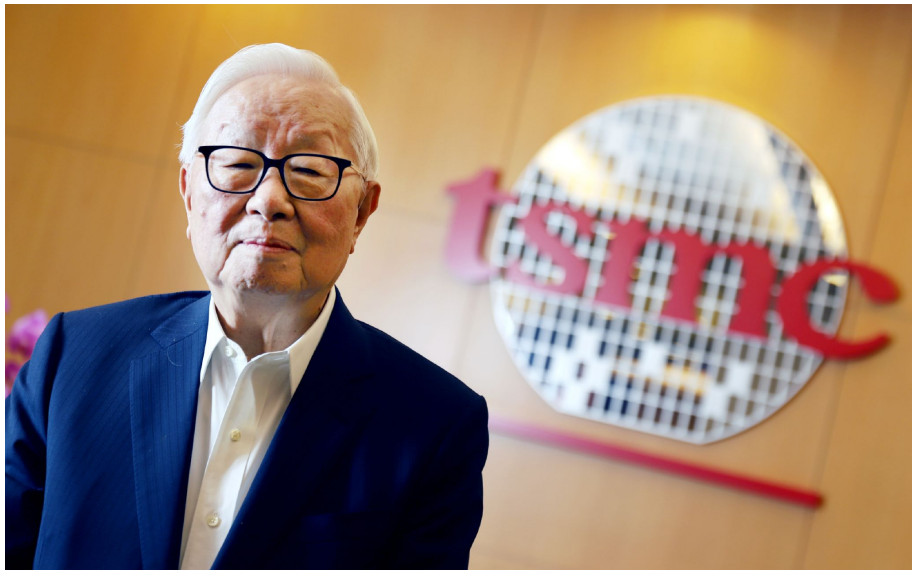
Cheaper Chips Made in Asia



Fairchild Semiconductors in Hong Kong. 1960s



1980s U.S. Manufacturers invest in Taiwan, S. Korea



TSMC Morris Chang



Samsung Lee Byung Chul

SUPERCHARGED

A $40,000 Nvidia chip has become the world's most sought-after hardware

Companies and governments want to deploy generative AI—but first they need access to Nvidia's H100 chips

By **Michelle Cheng** Published August 18, 2023



Nvidia is the AI company of the moment. Photo: Ann Wang (Reuters)

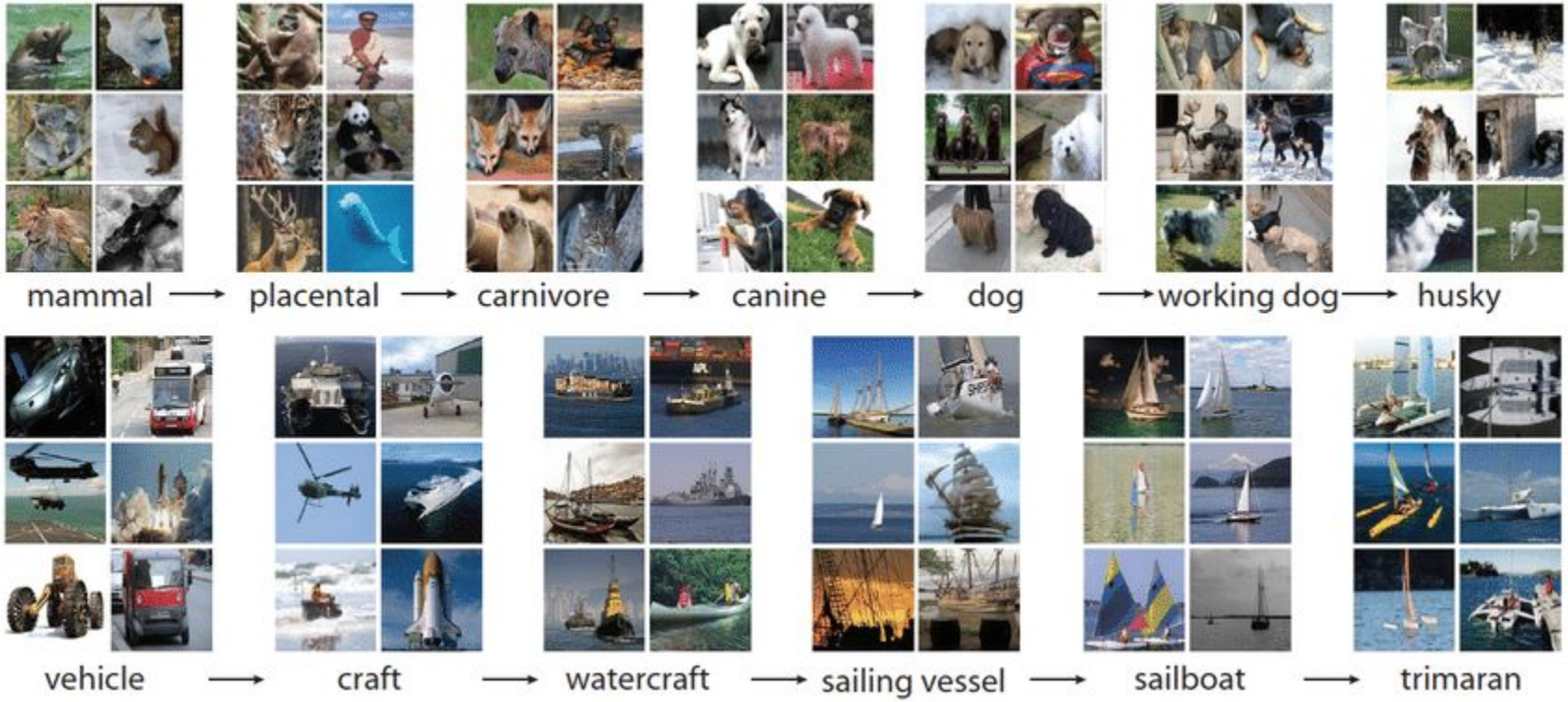
It seems like everyone wants to get their hands on Nvidia's H100 chips these days.

Microsoft and Google, which are building out generative AI-driven search engines, are some of the biggest customers of Nvidia's H100 chips. Server manufacturers claim to have waited more than six months to receive their latest orders. Venture capital investors are buying up H100 chips for the startups they invest in.

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Hardware to Software to Hardware(?)

Private to Public to Private Funds

Problem-Centered to Knowledge-Centered

Rule-Based to Connectionist

What is AI anyway?

Early Modern



Modern



Large-Scale AI?



- How do humans imagine themselves in relation to AI?
- What does this say about us?
- Who funds it? What motivates its development?
- Why do we keep making it?